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MSC RESEARCH PROJECT

Galois Cohomology and Local Class Field Theory

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Abstract

The goal of this project is to provide an accessible guide to the two main theorems of local class field theory. With these we establish a bijective correspondence between abelian extensions of a local field K and closed subgroups of K^* with finite index. It follows closely the work of Serre [1][2] but with the inclusion of some additional lemmas, proofs, and details omitted there. In addition, there are worked examples that should help the reader see the application of the two main theorems.

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Plagiarism statement

The work contained in this thesis is my own work unless otherwise stated.

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Chapter 1

Introduction

Classification plays an important role in mathematics. It allows one to understand the structure and big picture of what is going on. Local class field theory is no exception; we aim to classify abelian extensions of local fields. This is an important area as it builds an understanding of some fundamental objects in number theory. It ties together Galois theory and cohomological algebra to understand the cohomological structure of these fields and the isomorphisms that can arise.

This project is grouped in two main parts: the background chapter and the main results. The background chapter gives a brief overview of some of the objects we are studying before diving head-first into some abstract group cohomology. The group cohomology developed is instrumental in the main theorems we develop. The main results chapter sets us in the realm of field extensions and works to apply the group cohomology facts to this setting. The project as a whole loosely follows Local Fields by Serre [1], but with a personal spin via rearrangement and the author's proofs which include details and lemmas omitted in the text. Often many steps are skipped in the book, or proofs completely omitted in favour of a simple "Clear". The definitions and many of the statements of theorems are paraphrased from Local Fields. The goal of this work is to give a more accessible guide to the main theorems of local class field theory to students of a similar level. The proofs are done by the author, and where indicated, follow the ideas of a reference. It is hoped that future students will find this presentation helpful in avoiding some of the difficulties encountered by the author and are able to use their project to extend the ideas of Galois cohomology and class field theory to greater heights.

The project assumes the background knowledge of a master's student. In particular, it assumes familiarity with Galois theory, algebraic number theory, and some basics of homological algebra and category theory.

To give some motivation and idea of what we are looking at, we start with the simplest example of a local field extension: quadratic extensions of $\mathbb{Q}_p$ for p odd. This has been worked through before in the preliminary report [3] and serves as a good example to show what the main theorems we prove here are capable of. Quadratic extensions of $\mathbb{Q}_p$ are of the form $\mathbb{Q}_p(\sqrt{d})$ where d is a squarefree p-adic number. So to find such extensions, we look for non-

trivial elements in $\mathbb{Q}_p^*/\mathbb{Q}_p^{*2}$.

$$\begin{aligned}\mathbb{Q}_p^* &\cong \langle p\rangle \times \mathbb{Z}_p^* \\ \Rightarrow \mathbb{Q}_p^*/\mathbb{Q}_p^{*2} &\cong \langle p\rangle/\langle p\rangle^2 \times \mathbb{Z}_p^*/\mathbb{Z}_p^{*2} \\ &\cong \{1,p\} \times \mathbb{Z}_p^*/\mathbb{Z}_p^{*2}\end{aligned}$$

We just need to find elements of $\mathbb{Z}_p^*/\mathbb{Z}_p^{*2}$. The kernel of $\mathbb{Z}_p^* \to \mathbb{F}_p^*$ is $1+p\mathbb{Z}_p$. From this, one can deduce $\mathbb{F}_p^* \cong \mathbb{Z}_p^*/(1+p\mathbb{Z}_p)$. A computation using Hensel's lemma shows every $a \in 1+p\mathbb{Z}_p$ is a square. So we see:

$$\begin{aligned}\mathbb{Z}_p^*/\mathbb{Z}_p^{*2} &\cong (\mathbb{Z}_p^*/(1+p\mathbb{Z}_p))/\mathbb{Z}_p^{*2} \\ &\cong \mathbb{F}_p^*/\mathbb{F}_p^{*2} \\ &\cong \mathbb{Z}/2\mathbb{Z}\end{aligned}$$

The quotient then is equal to $\{1,u\}$ where u is some unit non-square in $\mathbb{Z}_p$. We find we have the extensions corresponding to $d = \{1,p,u,up\}$. $d=1$ is a trivial extension and so we are left with $d = \{p,u,up\}$. We did not need the main theorems here, but this example serves as a good check that the theorems work as intended.

Theorem 1 (Local Artin Map)**.** Let F/E be a Galois extension. Let $N : F \to E$ denote the norm map. Then there is an isomorphism:

$$\mathrm{Gal}(F/E)^{ab} \longrightarrow E^*/NF^*$$

[1]

Theorem 2. For a subgroup of E^* to be a norm group (and thus correspond to an abelian extension), it is necessary and sufficient that it be closed and finite index [1].

These together show a 1-to-1 correspondence between abelian Galois extensions over a local field E and subgroups of the group of units of E which are closed of finite index. We use these to explore the example we just worked out. Since every quadratic extension is abelian, finding all abelian extensions is sufficient. We now search for all closed index 2 subgroups of $\mathbb{Q}_p^*$. The subgroups must contain $\mathbb{Q}_p^{*2}$, so we simply look for order 2 subgroups of $\mathbb{Q}_p^*/\mathbb{Q}_p^{*2}$. By computations above, these are exactly:

1. $\langle (1,u)\rangle$
2. $\langle (p,1)\rangle$
3. $\langle (p,u)\rangle$

The result is the same, but the method is much more general and can be applied to many fields in which the computations we did above are not so easy. The rest of the project will build up the results needed to prove these theorems and then work through some examples. By the end hopefully one can understand the way they are applied and the way local field extensions work.

Chapter 2

Background

2.1 Valuations and Local Fields

We begin with some commutative algebra background to familiarize ourselves with the objects we are studying.

Definition 2.1.0.1 (Discrete valuation ring (DVR))**.** A discrete valuation ring, A, is a principal ideal domain which contains a unique non-zero prime ideal $\mathfrak{m}$.

We note that in a PID A, any ideal is given by $(\pi) = \pi A$ where $\pi \in A$.

Definition 2.1.0.2 (Uniformizer)**.** Let A be a DVR and $\pi \in A$ the irreducible element such that $\mathfrak{m} = (\pi)$. We call π a uniformizer.

We note that such a π is not unique; for a uniformizer π and a unit u, we see that $u\pi$ also generates $\mathfrak{m}$. It is unique up to multiplication by a unit.

Proof. Let π, π' generate $\mathfrak{m}$. Thus $\pi' = a\pi$ for some $a \in A$. Since $\mathfrak{m}$ is prime, π, π' are irreducible and we conclude a is a unit. □

Definition 2.1.0.3 (Residue field)**.** The field given by the quotient $A/\mathfrak{m}$ is called the residue field. [1]

Now since DVRs have unique prime (and thus maximal) ideals, we see they are local rings. Thus every element is either a unit or contained in the prime ideal. We can use this to write every element x as $u\pi^n$ for a chosen uniformizer π, some unit u and some $n \in \mathbb{N} \cup \{0\}$.

Definition 2.1.0.4 (Valuation)**.** Let A be a DVR and $a = u\pi^n \in A$. We call n the valuation of a and denote it $v(a)$. [1]

It is clear that the valuation is independent of one's choice of uniformizer as we have seen that uniformizers are unique up to multiplication by units. The valuation extends to the field of fractions of a DVR by allowing n to be any integer. The convention is that $v(0) = +\infty$.

Proposition 1. Let A be a DVR with K its field of fractions.

1. $v : K^* \to \mathbb{Z}$ is a surjective group homomorphism.
2. $v(x + y) \geq inf(x, y)$

Proof. We begin with 1. We saw above that the choice of uniformizer does not affect the valuation and thus the map is well defined. The surjection is easy to see by taking $v(\pi^z)$. Finally it is a homomorphism since $v(xy) = v(u\pi^n w\pi^m) = v(uw\pi^{m+n}) = m + n = v(x) + v(y)$.

Now 2. WLOG let $n \leq m$. Then
$v(x+y) = v(u\pi^n + w\pi^m) = v(\pi^n(u + w\pi^{m-n})) = n + v(u + w\pi^{m-n})$ Since $m - n \geq 0$, $u + w\pi^{m-n} \in A$ and its valuation must be non-negative. Thus $v(x+y) \geq n$. □

We can use the valuation to define a non-Archimedean absolute value by setting $|a| = |\pi|^{-v(a)}$ where π is a uniformizer.

Definition 2.1.0.5. The ring of integers A_K of a field K complete with respect to a non-Archimedean absolute value is the subset of K which has discrete valuation greater than or equal to 1. It is a discrete valuation ring by definition.

Definition 2.1.0.6 (Local Field)**.** We call a field K a local field if it is complete with respect to an Archimedean absolute value and has finite residue field.

Example 1. $\mathbb{Q}_p$ is a local field

Example 2. $\mathbb{R}$, $\mathbb{C}$ are local fields

Example 3. $\mathbb{Q}$ or any finite algebraic extension of $\mathbb{Q}$ is not a local field.

For a local field K we have that A_K is compact and K is locally compact.

2.2 Ramification Groups

In this section we let K be a local field. Let A_K be the ring of integers of K with unique prime ideal $\mathfrak{p}_K$, $\bar{K} = A_K/\mathfrak{p}_K$ the residue field and U_K the group of units of A_K. For a finite separable extension L/K we will use analogous notation to above for L. We also assume $\bar{L}/\bar{K}$ is separable. In this section we will simply quote results from [1] to understand the basics of ramification groups and the filtration on the group of units. Finally let v_K be the valuation which extends to v_L.

Proposition 2. Let L/K be a Galois extension with Galois group G. Let $x \in A_L$ be an element which generates A_L as a A_K algebra. Let $s \in G$ The following are equivalent:

1. s acts trivially on $A_L/\mathfrak{p}_L^{(i+1)}$
2. $v_L(s(a) - a) \geq i + 1$ for all $a \in A_L$
3. $v_L(s(x) - x) \geq i + 1$

Definition 2.2.0.1. Let $s \in G$ satisfying the equivalent conditions above for some $i \geq -1$. Then we call this subgroup the ith ramification group and denote it G_i.

Proposition 3. The G_i form a decreasing sequence of subgroups with $G_{-1} = G$ and $G_i = 1$ for i large enough.

Definition 2.2.0.2. The group G_0 is called the inertia subgroup of G.

The inertia subgroup has a definition via Galois theory as well but it is convenient here to define it as the ramification group.

Proposition 4. Let $s \in G_0$. Then $s \in G_i$ iff $s(\pi)/\pi \equiv 1 \bmod \mathfrak{p}_L^i$

Define the filtration of U_L:

$$U_L^{(0)} = U_L$$
$$U_L^{(i)} = 1 + \mathfrak{p}_L^i$$

This is a decreasing sequence of closed subgroups which form a neighbourhood of 1. The group U_L is closed in a complete DVR, hence is complete itself and we get:

$$U_L = \varprojlim U_L/U_L^{(i)}$$

Proposition 5. $U_L^{(0)}/U_L^{(1)} \cong \bar{L}^*$

This follows after one realizes the identity in $\bar{L}^*$ is the coset $1 + \mathfrak{p}_L$. For the following definitions let L/K be a Galois extension of local fields with Galois group G and inertia group G_0. Let v_K and v_L denote the valuations and π_k, π_L be uniformizers.

Definition 2.2.0.3. L/K is unramified if $v_K(\pi_L) = 1$ and ramified if $v_K(\pi_L) > 1$.

Definition 2.2.0.4. L/K is tamely ramified if it is ramified and $char(K) \nmid |G_0|$.

Definition 2.2.0.5. L/K is wildly ramified if it is ramified and $char(K) \mid |G_0|$

Definition 2.2.0.6. L/K is totally ramified if $v_K(\pi_L) = [L : K]$.

Definition 2.2.0.7. Let K be a local field with maximal ideal $\mathfrak{p}$. An Eisenstein polynomial is a non-zero polynomial $x^n + a_{n-1}x^{n-1} + \cdots + a_0$ such that $a_i \in \mathfrak{p}$, $a_0 \notin \mathfrak{p}^2$.

Theorem 3. If L/K is totally ramified then $L = K[\alpha]$ where α is the root of an Eisenstein polynomial [1].

2.3 Group Cohomology

We now develop some tools in group cohomology with the aim of applying them to Galois cohomology. Much of the heavy lifting is done in this section.

2.3.1 G-Modules and Group Cohomology

Definition 2.3.1.1 (Group ring)**.** The group ring $\mathbb{Z}[G]$ of G is a free $\mathbb{Z}$-module with generators $g \in G$.

Note that $g \in G$ acts on $\mathbb{Z}[G]$ by sending the generator h to gh.

Definition 2.3.1.2 (G-module)**.** Let G be a group written with multiplication as the action. A G-module is simply a $\mathbb{Z}[G]$-module with multiplication given by the action above.

We call any homomorphism of G-modules a G-homomorphism.

Proposition 6. For any G-module A, let A^G be the submodule of A which is fixed by the action of G on A. Any G-homomorphism $f : A \to B$ maps A^G to B^G.

Proof. Let $a \in A^G$. Then $\forall g \in G$, $ga = a$

$$\Rightarrow f(ga) = f(a)$$
$$\Rightarrow gf(a) = f(a)$$

and we see that $f(a)$ lies in B^G. □

Now we see that (-)^G behaves functorially.

Proposition 7. (-)^G is left-exact.

Proof. Let

$$0 \longrightarrow A \longrightarrow B \longrightarrow C$$

be exact. $A \to B$ restricted to A^G is still injective and lands in B^G by the proposition above. Thus $A^G \to B^G$ is injective. Now let $b \in \ker(B^G \to C^G)$. Then $b \in \mathrm{im}(A \to B)$ by exactness. Furthermore we know $b \in im(A^G \to B^G)$ by the proposition. Hence $\ker(B^G \to C^G) \subseteq \mathrm{im}(A^G \to B^G)$. Similarly if $b \in \mathrm{im}(A^G \to B^G)$ then $b \in \ker(B \to C)$ by exactness and $b \in B^G$. We see $b \in \ker(B^G \to C^G)$ and conclude $\ker(B^G \to C^G) = \mathrm{im}(A^G \to B^G)$. Hence

$$0 \longrightarrow A^G \longrightarrow B^G \longrightarrow C^G$$

is exact as well. □

(-)^G can be identified with $\mathrm{Hom}_G(\mathbb{Z}, -)$ which is the set of $\mathbb{Z}[G]$-morphisms from $\mathbb{Z}$ to (-). We use this to compute group cohomology. The nth cohomology of G with respect to a G-module A is denoted $\mathrm{H}^n(G, A)$ and is defined to be the nth derived functor of $\mathrm{Hom}_G(\mathbb{Z}, -)$. This yields $\mathrm{H}^0(G, A)$ immediately: $\mathrm{H}^0(G, A) = \mathrm{Hom}_G(\mathbb{Z}, A) = A^G$. To compute higher cohomology we can take an injective coresolution of A or a projective resolution $P_\bullet$ of $\mathbb{Z}$ by G-modules.

Definition 2.3.1.3 (Resolution)**.** A resolution of an R-module M is a complex $P_\bullet$ of R-modules P_i such that the following complex is exact:

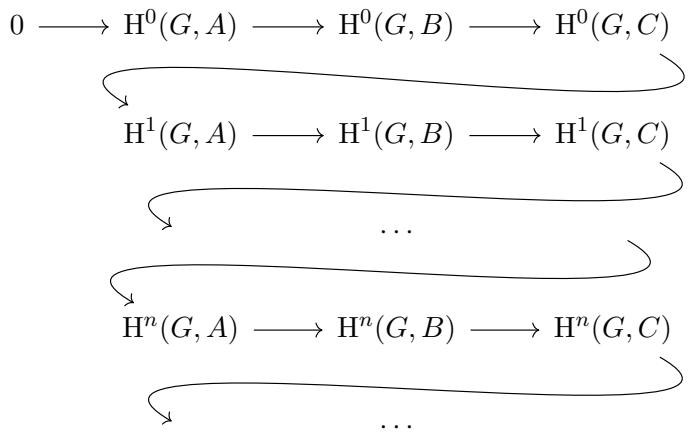
$$\ldots \longrightarrow P_2 \longrightarrow P_1 \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

A coresolution is the dual notion to a resolution. A projective resolution of is a resolution such that every $P_i \in P_\bullet$ is a projective module. An injective (co)resolution is a coresolution such that every $P_i \in P^\bullet$ is an injective module. If we have found a projective resoltuion, we take $\mathrm{Hom}_G(P_\bullet, A)$. $\mathrm{Hom}_G(-, A)$ is a contravariant functor from modules to abelian groups, giving us a cocomplex. If we have found an injective resolution (which is often harder) we take $\mathrm{Hom}_G(\mathbb{Z}, P^\bullet)$. This is covariant, giving us again a cocomplex. In either case, let us denote the cocomplex $Q^\bullet$. The cohomology group $\mathrm{H}^n(G, A)$ is the quotient $\ker(Q^n \to Q^{n+1})/\,\mathrm{im}(Q^{n-1} \to Q^n)$ and does not depend on the complex chosen.

Proposition 8. Any short exact sequence of G-modules:

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

induces a long exact sequence of cohomology:



The proof can be found in Weibel [4]. The curved arrow mapping $\mathrm{H}^n(G, C)$ to $\mathrm{H}^{n+1}(G, A)$ is called the connecting homomorphism. This is often denoted δ. The dual notion for homology also holds.

Definition 2.3.1.4 (Injective module)**.** An injective module A is a module such that any short exact sequence:

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

splits as:

$$0 \longrightarrow A \longrightarrow A \oplus B \longrightarrow C \longrightarrow 0$$

In particular, $B \cong A \oplus B$.

Proposition 9. If A is an injective module then $\mathrm{H}^n(G, A) = 0 \; \forall n \geq 1$.

Proof. Let A be an injective $\mathbb{Z}[G]$-module. Naturally, the following sequence is exact:

$$0 \longrightarrow A \longrightarrow A \longrightarrow 0$$

which shows that $Q^\bullet :=$

$$0 \longrightarrow A \longrightarrow 0$$

is an injective resolution of A by $\mathbb{Z}[G]$-modules. Taking $\mathrm{Hom}_G(\mathbb{Z}, Q^\bullet)$, we see that the only non-trivial cohomology is $\mathrm{H}^0(G, A)$. □

We define the group homology in a similar (dual) way to cohomology. They are the left derived functors of $\mathbb{Z} \otimes_{\mathbb{Z}[G]} -$ or simply $\mathbb{Z} \otimes_G -$. It is characterized by the following properties:

Proposition 10. Let $DA := \{sa - a : s \in G, a \in A\}$. Then:

1. $\mathrm{H}_0(G, A) = A/DA$
2. If A is projective then $\mathrm{H}_n(G, A) = 0 \; \forall n \geq 1$

Proof. For 1, we use the standard projective (free) resolution seen in Weibel [4]:

$$\cdots \longrightarrow \mathbb{Z}[G^2] \otimes_G A \longrightarrow \mathbb{Z}[G] \otimes_G A \longrightarrow 0$$

Since in general we have $R \otimes_R M \cong M$, we just aim to show $\mathrm{im}(\mathbb{Z}[G^2] \otimes_G A \to \mathbb{Z}[G] \otimes_G A) = DA$. Computing the image via cocycles, we see an elementary tensor $(g, h) \otimes a$ gets sent to $(h - g)a$ which is in DA. We see $\mathrm{im}(\mathbb{Z}[G^2] \otimes_G A \to \mathbb{Z}[G] \otimes_G A) \subseteq DA$. Now let $sa - a$ be

arbitrary in DA. Take $(e,s)\otimes a \in \mathbb{Z}[G^2]\otimes_G A$ (where e is the identity in G). This gets mapped to $(s-e)a = sa - a$. Hence $\mathbb{Z}[G]\otimes_G A) = DA$ and we have our result.

For 2, we note that when A is projective

$$0 \longrightarrow A \longrightarrow 0$$

is a projective resolution of A which gives the result. □

We denote $A_G := A/DA$

Definition 2.3.1.5. The map $\sigma : \mathbb{Z}[G] \to \mathbb{Z}$ given by $\sum n_i g_i \mapsto \sum n_i$ (where $n_i \in \mathbb{Z}$ and $g_i \in G$) is call the augmentation map. The kernel of σ, I_G, is called the augmentation ideal.

Note that $\mathbb{Z}[G]/I_G = \mathbb{Z}[G]_G$. We now compute an important homology group.

Example 4. $\mathrm{H}_1(G,\mathbb{Z}) = G^{ab}$. Consider the following exact sequence:

$$0 \longrightarrow I_G \xrightarrow{\iota} \mathbb{Z}[G] \xrightarrow{\sigma} \mathbb{Z} \longrightarrow 0$$

The induced long exact sequence on homology is the following:

$$\cdots \longrightarrow \mathrm{H}_1(G,I_G) \longrightarrow \mathrm{H}_1(G,\mathbb{Z}[G]) \longrightarrow \mathrm{H}_1(G,\mathbb{Z})$$
$$\mathrm{H}_0(G,I_G) \longrightarrow \mathrm{H}_0(G,\mathbb{Z}[G]) \longrightarrow \mathrm{H}_0(G,\mathbb{Z}) \longrightarrow 0$$

Since $\mathbb{Z}[G]$ is projective (free), $\mathrm{H}_1(G,\mathbb{Z}[G]) = 0$. In addition, $\mathrm{H}_0(G,I_G) = I_G/I_G^2$. The inclusion of this into $\mathrm{H}_0(G,\mathbb{Z}[G]) = \mathbb{Z}[G]/I_G$ is zero. Therefore we pull the following short exact sequence out of the long exact sequence:

$$0 \longrightarrow \mathrm{H}_1(G,\mathbb{Z}) \longrightarrow \mathrm{H}_0(G,I_G) \longrightarrow 0$$

Which implies $\mathrm{H}_1(G,\mathbb{Z}) \cong \mathrm{H}_0(G,I_G) = I_G/I_G^2$. Finally one can verify that $\psi : G \to I_G/I_G^2$ given by $s \mapsto s-1$ defines by passage to the quotient an isomorphism of G^{ab} to I_G/I_G^2 [5]. We conclude that $\mathrm{H}_1(G,\mathbb{Z}) = G^{ab}$.

Under certain conditions, we can find a homomorphism between $\mathrm{H}^n(G,A)$ and $\mathrm{H}^n(G',A')$. A group homomorphism $\varphi : G' \to G$ induces a homomorphism $\varphi^* : \mathrm{H}^n(G,A) \to \mathrm{H}^n(G',A)$. In a similar way, a G-module homomorphism $f : A \to A'$ induces a homomorphism $f^* : \mathrm{H}^n(G,A) \to \mathrm{H}^n(G,A')$. These are called compatible if $g' \cdot f(a) = f(\varphi(g')\cdot a)$. Some important examples of include

1. H a subgroup of G. We obtain the *restriction* homomorphisms $\mathrm{Res} : \mathrm{H}^n(G,A) \to \mathrm{H}^n(H,A)$ induced by the inclusion of H into G.
2. H is a normal subgroup of G. Then A^H is a G/H-module and we obtain the *inflation* homomorphisms $\mathrm{inf} : \mathrm{H}^n(G/H,A^H) \to \mathrm{H}^n(G,A)$.

Similar results hold for homology, but we are mostly interested in the cohomological case. This is due to the fact that we have an interesting exact sequence:

$$0 \longrightarrow \mathrm{H}^1(G/H, A^H) \xrightarrow{\mathrm{Inf}} \mathrm{H}^1(G, A) \xrightarrow{\mathrm{Res}} \mathrm{H}^1(H, A)$$

Proposition 11. The above sequence is exact.

Proof. We prove exactness at each cohomology group.

1. We begin with exactness at $\mathrm{H}^1(G/H, A^H)$. Let c be a 1-cocycle in $\mathrm{H}^1(G/H, A^H)$ which maps to a trivial 1-cocycle in $\mathrm{H}^1(G, A)$. By considering c as a class function on G, we can evaluate c on all of G. We have $c(e) = c(e \cdot e) = ec(e) + c(e) = 2c(e)$ (where e is the identity in G). This implies $c(e) = 0$ and thus $c(H) = 0$ (since H is the identity class). Now, if c is trivial in $\mathrm{H}^1(G, A)$, then there exists $m \in M$ such that $c(g) = gm - m$. Then $\forall h \in H$ we have $0 = c(h) = hm - m$. We see $m \in M^H$ and so has the form $c(gH) = gHm - m$ in $\mathrm{H}^1(G/H, A^H)$. Thus c is trivial in $\mathrm{H}^1(G/H, A^H)$. We conclude that Inf is injective.

2. We now check exactness at $\mathrm{H}^1(G, A)$. First, we verify that $\mathrm{Res} \circ \mathrm{Inf} = 0$. Let c be a cocycle in the image of Inf. Then c acts as a class function on G with classes being the cosets in G/H. By the same argument as in 1, $c(H) = 0$. Restricting this to H via the restriction map shows $\mathrm{Res} \circ \mathrm{Inf} = 0$. Now let c be a 1-cocycle in $\mathrm{H}^1(G, A)$ such that its restriction to H is trivial. In particular, we have $\exists m \in M$ such that $\forall h \in H$, $c(h) = hm - m$. Let $c'(g) = c(g) - (gm - m)$. We see that c, c' lie in the same class. Now $c'(h) = 0$ for any $h \in H$ thus $c'(gh) = gc'(h) + c'(g) = c'(g)$. This shows that c' is constant on G/H cosets and so we can consider it as a map $G/H \to M$. Finally, since H is normal in G we have $c'(g) = c'(gh) = c'(hg) = hc'(g) + c'(h) = hc'(g)$ which shows $c'(g) \in M^H$. We conclude that c' lies in the image of Inf.

□

This can be extended to the following proposition.

Proposition 12. Let $n \geq 1$ and suppose that for every $1 \leq i \leq n-1$, $\mathrm{H}^i(H, A) = 0$. Then

$$0 \longrightarrow \mathrm{H}^n(G/H, A^H) \xrightarrow{\mathrm{Inf}} \mathrm{H}^n(G, A) \xrightarrow{\mathrm{Res}} \mathrm{H}^n(H, A)$$

is exact

The proof is found in Weibel [4]. Let $[G : H] = m < \infty$. We now define maps $\mathrm{Cor} : \mathrm{H}^n(H, A) \to \mathrm{H}^n(G, A)$ in the other direction called corestriction. When $n = 0$ we define $N_{G/H}(a) = \sum_{s \in G/H} sa$ for $a \in A^H$. Since G/H is finite, it makes sense to make this sum. We call this the *norm* of a. It is clear that this lands in A^G since $g \in G$ acts on the $s \in G/H$ by permutation. It extends to a morphism of the cohomological functor $\mathrm{H}^n(G, -)$ [4].

Proposition 13. Let $m = |G/H|$. Then the composition $A^G \xrightarrow{\mathrm{Res}} A^H \xrightarrow{\mathrm{Cor}} A^G$ is equivalent to the multiply by m map.

Proof. Let $a \in A^G$. Then $\mathrm{Res}(a) = a \in A^H$. $N_{G/H}(a) = \sum_{s \in G/H} sa$. However, $a \in A^G$ so $\sum_{s \in G/H} sa = \sum_{s \in G/H} a = ma$. □

Theorem 4. Let $m = |G/H|$. Then the composition $\mathrm{H}^n(G, A) \xrightarrow{\mathrm{Res}} \mathrm{H}^n(H, A) \xrightarrow{\mathrm{Cor}} \mathrm{H}^n(G, A)$ is equivalent to the multiply by m map.

This is an extension of the previous proposition and we have dual results for homology [4].

Definition 2.3.1.6. Let G be a finite group. The norm of $\mathbb{Z}[G]$, denoted N, is equal to $\sum_{s\in G} s$.

For any G-module A, multiplication by N defines an endomorphism of A. It is also denoted N.

Proposition 14. $I_G A \subset \ker(\mathrm{N})$ and $\mathrm{im}(\mathrm{N}) \subset A^G$.

Proof. We first note that I_G has $\mathbb{Z}$ basis $1-s$ as s runs through G. On any of these generators we see

$$\begin{aligned}\mathrm{N}(1-t) &= \mathrm{N}(1) - \mathrm{N}(\mathrm{t}) \\ &= \sum_{s\in G} s - \sum_{s\in G} ts \\ &= \sum_{s\in G} s - \sum_{s'\in G} s' \\ &= \mathrm{N}(1) - \mathrm{N}(1) = 0\end{aligned}$$

Now let $a \in \mathrm{im}(\mathrm{N})$. Then $a = \sum_{s\in G} sa'$ and $ga = \sum_{s\in G} gsa' = \sum_{s'\in G} s'a' = a$. □

Thus N induces a homomorphism:

$$\mathrm{N}^* : A/I_G A \to A^G$$

One might recognize $A/I_G A = \mathrm{H}_0(G, A)$, $A^G = \mathrm{H}^0(G, A)$.

Definition 2.3.1.7. The Tate cohomological functor is defined:

$$\begin{aligned}\hat{\mathrm{H}}_0(G, A) &= \ker(\mathrm{N}^*) \\ \hat{\mathrm{H}}^0(G, A) &= \mathrm{coker}(\mathrm{N}^*)\end{aligned}$$

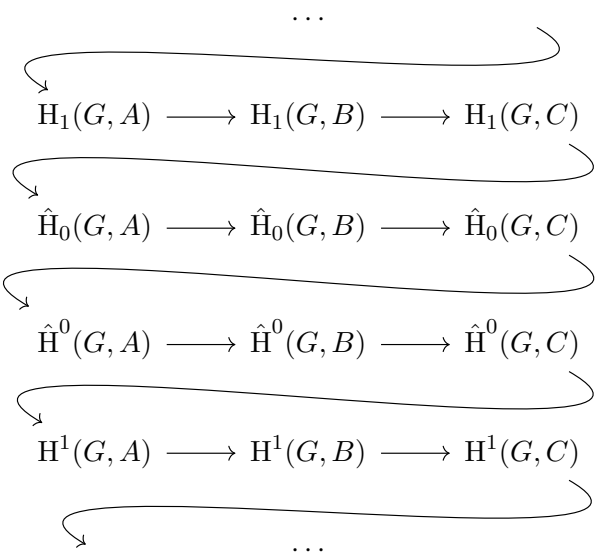
Now let

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be exact. Then one can check that the following diagram commutes

$$\begin{array}{ccccccccc}
\mathrm{H}_1(G,C) & \longrightarrow & \mathrm{H}_0(G,A) & \longrightarrow & \mathrm{H}_0(G,B) & \longrightarrow & \mathrm{H}_0(G,C) & \longrightarrow & 0 \\
\downarrow & & \downarrow{\scriptstyle \mathrm{N}^*_{\mathrm{A}}} & & \downarrow{\scriptstyle \mathrm{N}^*_{\mathrm{B}}} & & \downarrow{\scriptstyle \mathrm{N}^*_{\mathrm{C}}} & & \downarrow \\
0 & \longrightarrow & \mathrm{H}^0(G,A) & \longrightarrow & \mathrm{H}^0(G,B) & \longrightarrow & \mathrm{H}^0(G,C) & \longrightarrow & \mathrm{H}^1(G,A)
\end{array}$$

This diagram defines a canonical connecting homomorphism $\delta : \ker(\mathrm{N}^*_{\mathrm{C}}) \to \mathrm{coker}(\mathrm{N}^*_{\mathrm{A}})$[5]. Note that this is in fact a homomorphism $\delta : \hat{\mathrm{H}}_0(G, C) \to \hat{\mathrm{H}}^0(G, A)$. In fact such a diagram gives an exact sequence [5]:



This leads us to define the Tate cohomology groups.

Definition 2.3.1.8. The Tate cohomology groups are given by:

$$\begin{aligned}
\hat{\mathrm{H}}^n(G,A) &= \mathrm{H}^n(G,A),\ n \geq 1 \\
\hat{\mathrm{H}}^0(G,A) &= \mathrm{coker}(\mathrm{N}^*) = A^G/NA \\
\hat{\mathrm{H}}^{-1}(G,A) &= \ker(\mathrm{N}^*) = {}_N A/I_G A \\
\hat{\mathrm{H}}^{-n}(G,A) &= \mathrm{H}_{n-1}(G,A),\ n \geq 2
\end{aligned}$$

Where ${}_N A$ is the kernel on N acting on A.

Definition 2.3.1.9 (Induced Module)**.** Let A be a G-module. A is said to be induced if $A \cong \mathbb{Z}[G] \otimes_{\mathbb{Z}} X$ where X is an abelian group.

Definition 2.3.1.10 (Relatively Projective)**.** A G-module A is said to be relatively projective if it is a direct summand of an induced module.

Theorem 5. Let $m = [G : H]$. Then the composition $\hat{\mathrm{H}}^n(G,A) \xrightarrow{\mathrm{Res}} \hat{\mathrm{H}}^n(H,A) \xrightarrow{\mathrm{Cor}} \hat{\mathrm{H}}^n(G,A)$ is equivalent to the multiply by m map.

This follows from the already referenced the result for homology and cohomology 4 [4]. These cohomology groups are instrumental in proving one of our main theorems.

2.3.2 Theorems of Tate and Nakayama

The proofs in this subsection follow the ideas of Serre [1].

Definition 2.3.2.1. Let G be a finite group and A a G-module. A is called cohomologically trivial if $\hat{\mathrm{H}}^n(H,A) = 0$ for every subgroup H of G and for all $n \in \mathbb{Z}$.

Lemma 1. Let G be a p-group acting on a finite set S with S^G being the elements fixed by the G-action. Then $|S| \equiv |S^G| \bmod p$.

Proof. Consider $S \setminus S^G$. This is the disjoint union of orbits which are not a singular point. By the orbit-stabilizer theorem we have that the cardinality of each of these is equal to the index of their stabilizer in G. Since G is a p-group, each of these cardinalities divide p. □

Lemma 2. If a p-group acts on a non-trivial p-group, then the number of fixed points is strictly greater than 1.

Proof. Use lemma 1 and note that the number of fixed points is at least one (identity) and must divide p. □

Theorem 6. The center of a non-trivial p-group has cardinality strictly greater than 1.

Proof. Let G be a p-group. We use lemma 2 and let G act on itself by inner-automorphisms. □

Theorem 7. Every linear representation of a p-group over a field of characteristic p contains the unit representation.

Proof. Let S be the representation space. Let $0 \neq x \in S$. Let $H = \{xg \in S : g \in G\}$. We can view H as a finite dimensional vector space over $\mathbb{F}_p$ and thus a p-group. We have seen $|H^G| > 1$ (2), which shows there is a non-zero element in the fix of G. □

Theorem 8. Let G be a finite group. Let G_p be a Sylow-p subgroup. Then for every G-module A and every $n \in \mathbb{Z}$:

$$\mathrm{Res} : \hat{\mathrm{H}}^n(G, A) \to \hat{\mathrm{H}}^n(G_p, A)$$

is injective on the p-primary part of $\hat{\mathrm{H}}^n(G, A)$.

Proof. Let $x \in \ker(\mathrm{Res})$. Let $q = |G/G_p|$. Then theorem 5 shows $\mathrm{Cor} \circ \mathrm{Res} : \hat{\mathrm{H}}^n(G, A) \to \hat{\mathrm{H}}^n(G, A)$ factoring through $\hat{\mathrm{H}}^n(G_p, A)$ is the multiplication by q map. Thus $qx = 0$. Suppose such an x lies in the p-primary component of $\hat{\mathrm{H}}^n(G, A)$. Then $\exists r \in \mathbb{N}$ such that $p^r x = 0$. Now by the Sylow subgroup properties $(p^r, q) = 1$. Hence we conclude that $x = 0$ and we get the injectivity desired. □

Corollary 1. With the notation of the previous theorem, suppose that for every prime p we have some G_p such that $\hat{\mathrm{H}}^n(G_p, A) = 0$. Then $\hat{\mathrm{H}}^n(G, A) = 0$.

Proof. This follows from the previous theorem since every p-primary component is 0. □

Lemma 3. A relatively projective G-module is cohomologically trivial.

Found in Serre [1]. Note that this holds in particular for induced modules. We now introduce a little abstract nonsense to help us figure out when modules are cohomologically trivial.

Proposition 15. Let $\mathcal{C}$ be the category of abelian groups. Let $\mathrm{T} : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ be an additive bifunctor which is bicovariant. Let A, B be G-modules (hence also abelian groups). Let $s_A \in \mathrm{Hom}_{\mathbb{Z}}(A, A)$ and $s_B \in \mathrm{Hom}_{\mathbb{Z}}(B, B)$ be the elements given by multiplication by $s \in G$ respectively. Then we have an element $\mathrm{T}(s_A, s_B) \in \mathrm{Hom}(\mathrm{T}(A, B))$. This gives a G-action on $\mathrm{T}(A, B)$ making it a G-modules itself. If A is induced (respectively relatively projective) then $\mathrm{T}(A, B)$ is induced (respectively relatively projective).

Proof. We prove it for A induced as the relatively projective case follows by passing to a direct factor. So let A be an induced G-module. We can then write $A = \oplus_{s \in G} sA'$ for A' a subgroup of A. Then using functorial properties we see:

$$\begin{aligned} \mathrm{T}(A, B) &= \oplus_{s \in G} \mathrm{T}(sA', B) \\ &= \oplus_{s \in G} \mathrm{T}(sA', sB) \\ &= \oplus_{s \in G} s\,\mathrm{T}(A', B) \end{aligned}$$

Which shows that $\mathrm{T}(A, B)$ is induced. □

Note that $- \otimes_{\mathbb{Z}} -$, $\mathrm{Hom}_{\mathbb{Z}}(-,-)$, Tor and Ext all satisfy the conditions of T. We will use this later.

Lemma 4. Let G be a p-group and A a G-module such that $pA = 0$. Then the following are equivalent:

1. $A = 0$
2. $\mathrm{H}^0(G, A) = 0$
3. $\mathrm{H}_0(G, A) = 0$

Proof. $1 \Rightarrow 2$ and $1 \Rightarrow 3$ are trivial. Since $pA = 0$, A is a finite dimensional vector space over $\mathbb{F}_p$. Theorem 7 says that if A is non-zero then there exists a non-zero element of $y \in A$ such that $gy = y$ for all $g \in G$. Hence $A^G \neq 0$. The contrapositive of this shows $2 \Rightarrow 1$. Now let $A' = \mathrm{Hom}(A, \mathbb{F}_p)$ be the dual of A as a $\mathbb{F}_p$ vector space. Using some category theory, we can see $\mathrm{H}^0(G, A') = \mathrm{H}_0(G, A)^*$. Thus:

$$\begin{aligned} &\mathrm{H}_0(G, A) = 0 \\ \Rightarrow &\mathrm{H}^0(G, A') = 0 \\ \Rightarrow &A' = 0 \text{ (by } 2 \Rightarrow 1) \\ \Rightarrow &\mathrm{Hom}(A, \mathbb{F}_p) = 0 \\ \Rightarrow &A = 0 \text{ (since } A \text{ is an } \mathbb{F}_p \text{ vector space)} \end{aligned}$$

This shows $3 \Rightarrow 1$. □

Lemma 5. Keeping the hypotheses of the previous lemma, assume that $\mathrm{H}_1(G, A) = 0$. Then A is a free $\mathbb{F}_p[G]$-module.

Proof. Let $\mathfrak{r}$ be the augmentation ideal of $\mathbb{F}_p[G]$. Then, since $pA = 0$, $\mathrm{H}_0(G, A) = A/I_G A = A/\mathfrak{r}A$ where I_G is the augmentation ideal of $\mathbb{Z}[G]$. This makes $A/\mathfrak{r}A$ a vector space over $\mathbb{F}_p$. We choose a basis h_λ and lift to a family a_λ in A. We now want to show that a_λ generates A and to do so we will apply lemma 4 with $A/\langle a_\lambda \rangle$. We see:

$$\begin{aligned} \mathrm{H}_0(A/\langle a_\lambda \rangle, G) &= (A/\langle a_\lambda \rangle)/\mathfrak{r}(A/\langle a_\lambda \rangle) \\ &= A/\langle a_\lambda \rangle/(\mathfrak{r}A + \langle a_\lambda \rangle)/\langle a_\lambda \rangle \\ &= A/(\mathfrak{r}A + \langle a_\lambda \rangle) \end{aligned}$$

Now since a_λ contains coset representatives for all cosets in $A/\mathfrak{r}A$, we see that $A = \mathfrak{r}A + \langle a_\lambda \rangle$ which shows $\mathrm{H}_0(A/\langle a_\lambda \rangle, G) = 0$. By the previous lemma, $A/\langle a_\lambda \rangle = 0$ and thus $\langle a_\lambda \rangle$ generates A as an $\mathbb{F}_p[G]$-module. So we have a free $\mathbb{F}_p[G]$-module $\langle a_\lambda \rangle$ which surjects onto A. Also, by construction, we have an isomorphism $A/\mathfrak{r}A \cong \langle a_\lambda \rangle/\mathfrak{r}\langle a_\lambda \rangle$. Let $R = \ker(\langle a_\lambda \rangle \to A)$. Then we have the short exact sequence:

$$0 \longrightarrow R \longrightarrow L \longrightarrow A \longrightarrow 0$$

which induces a long exact sequence on homology and admits the following section:

$$\mathrm{H}_1(G, A) \longrightarrow \mathrm{H}_0(G, R) \longrightarrow \mathrm{H}_0(G, \langle a_\lambda \rangle) \longrightarrow \mathrm{H}_0(G, A)$$

but $\mathrm{H}_0(G, A) \cong A/\mathfrak{r}A \cong \langle a_\lambda \rangle/\mathfrak{r}\langle a_\lambda \rangle \cong \mathrm{H}_0(G, \langle a_\lambda \rangle)$. From this we see $\ker(\mathrm{H}_0(G, \langle a_\lambda \rangle) \to \mathrm{H}_0(G, A)) = 0$ and thus $\mathrm{im}(\mathrm{H}_0(G, R) \to \mathrm{H}_0(G, \langle a_\lambda \rangle)) = 0$. Also, $\mathrm{H}_1(G, A) = 0$ by assumption and we see $\mathrm{H}_0(G, R) = 0$. Again using lemma 4 we see $R = 0$. Hence $A \cong \langle a_\lambda \rangle$ and A is $\mathbb{F}_p[G]$-free. □

Theorem 9. Let G be a p-group and A a G-module such that $pA = 0$. Then the following are equivalent:

1. $\exists q \in \mathbb{Z}$ such that $\hat{\mathrm{H}}^q(G, A) = 0$
2. A is cohomologically trivial
3. A is an induced G-module
4. A is a free $\mathbb{F}_p[G]$-module

Proof. $4 \Rightarrow 3$ is because $4 \Rightarrow A \cong A \otimes_{\mathbb{Z}} \mathbb{F}_p$ which shows A is induced. We have seen $3 \Rightarrow 2$ already (lemma 3). $2 \Rightarrow 1$ is obvious. Thus it remains to show $1 \Rightarrow 4$. Since $pA = 0$, we can consider A as an $\mathbb{F}_p$ vector space. Hence A injects into a free $\mathbb{F}_p[G]$-module. Let $A_1 = F/A$. Since F is killed by p, so is A_1. We have the following exact sequence:

$$0 \longrightarrow A \longrightarrow F \longrightarrow A_1 \longrightarrow 0$$

However, by $3 \Rightarrow 2$ we know that F is cohomologically trivial. Hence the long exact sequence induced gives the short exact sequences:

$$0 \longrightarrow \hat{\mathrm{H}}^{n-1}(G, A_1) \longrightarrow \hat{\mathrm{H}}^n(G, A) \longrightarrow 0$$

Implying $\hat{\mathrm{H}}^{n-1}(G, A_1) \cong \hat{\mathrm{H}}^n(G, A)$. Then by repeating the process with the module A_1 we get the short exact sequences:

$$0 \longrightarrow \hat{\mathrm{H}}^{n-1}(G, A_2) \longrightarrow \hat{\mathrm{H}}^n(G, A_1) \longrightarrow 0$$

Implying that $\hat{\mathrm{H}}^{n-2}(G, A_2) \cong \hat{\mathrm{H}}^{n-1}(G, A_1) \cong \hat{\mathrm{H}}^n(G, A)$. by iterating this process, we get a G-module B such that $\hat{\mathrm{H}}^{n-q-2}(G, B) \cong \hat{\mathrm{H}}^n(G, A)$. By assumption, $\hat{\mathrm{H}}^n(G, A) = 0$. Thus $\hat{\mathrm{H}}^{-2}(G, B) = H_1(G, B) = 0$. By lemma 5, B is $\mathbb{F}_p[G]$-free. Hence, by $4 \Rightarrow 3 \Rightarrow 2$, we see B is cohomologically trivial. We see $\mathrm{H}_1(G, A) \cong \hat{\mathrm{H}}^{-2}(G, A) \cong \hat{\mathrm{H}}^{-q-4}(G, B) = 0$. Again by lemma 5, we conclude that A is $\mathbb{F}_p[G]$-free. □

Theorem 10. Let G be a p-group and A a G-module with no p-torsion. Then the following are equivalent:

1. $\hat{\mathrm{H}}^q(G, A) = 0$ for two consecutive values of q
2. A is cohomologically trivial
3. A/pA is a free $\mathbb{F}_p[G]$-module

Proof. Since A is p-torsion free, we have the following exact sequence:

$$0 \longrightarrow A \xrightarrow{p} A \longrightarrow A/pA \longrightarrow 0$$

$2 \Rightarrow 1$ is trivial. Now assuming 1 ($\hat{\mathrm{H}}^q(G, A) = \hat{\mathrm{H}}^{q+1}(G, A) = 0$), the long exact sequence of Tate cohomology shows that $\hat{\mathrm{H}}^q(G, A/pA) = 0$. Then theorem 9 says that A/pA is a free $\mathbb{F}_p[G]$-module giving us $1 \Rightarrow 3$. Assuming 3, theorem 9 says A/pA is cohomologically trivial. The long exact sequence shows $\hat{\mathrm{H}}^q(G, A) \xrightarrow{p} \hat{\mathrm{H}}^q(G, A)$ is bijective for all q. By composing, we see multiplication by p^n is bijective. Consider the map $\hat{\mathrm{H}}^q(G, A) \xrightarrow{\mathrm{Res}} \hat{\mathrm{H}}^q(\mathrm{id}, A) \xrightarrow{\mathrm{Cor}} \hat{\mathrm{H}}^q(G, A)$. From theorem 5 we know that this is equivalent to multiplication by $[G : \mathrm{id}] = |G| = p^n$. However, $\hat{\mathrm{H}}^q(\mathrm{id}, A) = 0$. Thus, multiplication by p^n acts as the zero map and is bijective on $\hat{\mathrm{H}}^q(G, A)$. We conclude $\hat{\mathrm{H}}^q(G, A) = 0$ giving us $3 \Rightarrow 2$. □

Corollary 2. Let A be a $\mathbb{Z}$-free G-module satisfying the equivalent conditions of the previous theorem. Then for every torsion-free module B, the G-module $N := \mathrm{Hom}_{\mathbb{Z}}(\mathrm{A}, \mathrm{B})$ is cohomologically trivial.

Proof. Since B is torsion-free we get the following exact sequence:

$$0 \longrightarrow B \xrightarrow{p} B \longrightarrow B/pB \longrightarrow 0$$

Since A is $\mathbb{Z}$-free, $\mathrm{Hom}_{\mathbb{Z}}(A, -)$ is exact and we get the exact sequence:

$$0 \longrightarrow N \xrightarrow{p} N \longrightarrow \mathrm{Hom}(A, B/pB) \longrightarrow 0$$

This shows an isomorphism $N/pN \cong \mathrm{Hom}(A, B/pB)$. Now, since B/pB is killed by p, any morphism $f \in \mathrm{Hom}(A, B/pB)$ has $0 = pf(a) = f(pa)$ which shows f is 0 on pA. Thus we have an isomorphism $N/pN \cong \mathrm{Hom}(A, B/pB) \cong \mathrm{Hom}(A/pA, B/pB)$. By assumtion (conditions of theorem 6), A/pA is $\mathbb{F}_p[G]$-free, hence induced. Proposition 15 ensures that N/pN is induced as well, hence cohomologically trivial. This and the long exact sequence of Tate cohomology induced by the above exact sequence gives the exact sequences:

$$0 \longrightarrow \hat{\mathrm{H}}^n(G, N) \xrightarrow{p} \hat{\mathrm{H}}^n(G, N) \longrightarrow 0$$

By the same argument as theorem 10 we see that N is cohomologically trivial. □

Theorem 11. Let G be a finite group, A a $\mathbb{Z}$-free G-module and G_p a Sylow p-subgroup of G for each p. The following are equivalent:

1. For every prime p, A satisfies the equivalent conditions of theorem 10 as a G_p-module
2. A is $\mathbb{Z}[G]$-projective.

Proof. Clearly if A is $\mathbb{Z}[G]$-projective, it is relatively projective and hence cohomologically trivial. It remains to prove $1 \Rightarrow 2$. Assume 1 and let $A = L/N$ where L is $\mathbb{Z}[G]$-free so the following sequence is exact:

$$0 \longrightarrow N \longrightarrow L \longrightarrow A \longrightarrow 0$$

Since A is $\mathbb{Z}$-free, we get the exact sequence:

$$0 \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(A, N) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(A, L) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(A, A) \longrightarrow 0$$

The corollary to theorem 10 along with our assumptions says that the G_p-module $\mathrm{Hom}_{\mathbb{Z}}(A, N)$ is cohomologically trivial for all p. By the corollary to theorem 8 we see that $\mathrm{H}^1(G, \mathrm{Hom}_{\mathbb{Z}}(A, N)) = 0$. Thus the long exact sequence induced by the above exact sequence has the following section:

$$0 \longrightarrow \mathrm{Hom}_G(A, L) \longrightarrow \mathrm{Hom}_G(A, A)$$

Showing that $\mathrm{Hom}_G(A, L)$ surjects onto $\mathrm{Hom}_G(A, A)$. Thus $A \xrightarrow{\mathrm{id}} A$ extends to a G-morphism $\varphi : L \to A$. Thus A is a direct factor of a free G-module and is $\mathbb{Z}[G]$-projective. □

Lemma 6. Let G be a finite group and

$$0 \longrightarrow A_1 \longrightarrow A_2 \longrightarrow \cdots \longrightarrow A_n \longrightarrow 0$$

be an exact sequence of G-modules such that all but one are cohomologically trivial. Then the last is cohomologically trivial as well.

Proof. Let $N_i = \ker(A_i \to A_{i+1})$, $N_0 = N_{n+1} = 0$. We obtain the following $n+1$ exact sequences (where $A_0 = 0$):

$$0 \longrightarrow N_i \longrightarrow A_i \longrightarrow N_{i+1} \longrightarrow 0$$

Label these S_i. By assumption, A_i is cohomologically trivial for all $i \neq q$. The long exact sequence of Tate cohomology induced by S_0 shows that $\hat{\mathrm{H}}^k(G, N_1) = 0 \; \forall k \in \mathbb{Z}$. If $q \neq 1$ then this in turn shows that N_2 is cohomologically trivial in the same way. This continues until we hit q. In a similar manner, if $q \neq n$ then the long exact sequence of Tate cohomology induced by S_n shows that N_n is cohomologically trivial. In turn, if $q \neq n-1$, we see N_{n-1} is cohomologically trivial. This also continues until we hit q. We see that N_i is cohomologically trivial for all i. We now turn to S_q. The long exact sequence induced by this admits the sections:

$$\hat{\mathrm{H}}^{k-1}(G, N_{q+1}) \longrightarrow \hat{\mathrm{H}}^k(G, N_q) \longrightarrow \hat{\mathrm{H}}^k(G, X_q) \longrightarrow \hat{\mathrm{H}}^k(G, N_{q+1})$$

However, the N_i are cohomologically trivial! We obtain the short exact sequences:

$$0 \longrightarrow 0 \longrightarrow \hat{\mathrm{H}}^k(G, X_q) \longrightarrow 0$$

Which shows $\hat{\mathrm{H}}^k(G, X_q) = 0 \; \forall k$ as needed. □

Theorem 12. Let G be a finite group, A a G-module. Let G_p be a Sylow p-subgroup of G. The following are equivalent:

1. For each prime p, $\exists q$ such that $\hat{\mathrm{H}}^q(G_p, A) = \hat{\mathrm{H}}^{q+1}(G_p, A) = 0$

2. A is cohomologically trivial

3. There exists an exact sequence $0 \to P_1 \to P_0 \to A \to 0$ where P_i is $\mathbb{Z}[G]$-projective.

Proof. Using lemma 6 shows $3 \Rightarrow 2$. $2 \Rightarrow 1$ immediately. It remains only to show $1 \Rightarrow 3$. Write A as a quotient of a $\mathbb{Z}[G]$-free module L so we get the exact sequence:

$$0 \longrightarrow N \longrightarrow L \longrightarrow A \longrightarrow 0$$

Since L is $\mathbb{Z}[G]$ free, it is $\mathbb{Z}$-free. As $\mathbb{Z}$ is a principal ideal domain, every submodule of a free module is free. Hence N is $\mathbb{Z}$-free and is a G_p-module with no p-torsion. Since L is $\mathbb{Z}[G]$-free and A satisfies condition 1, we see $\exists q$ such that $\hat{\mathrm{H}}^q(G_p, N) = \hat{\mathrm{H}}^{q+1}(G_p, N) = 0$. Hence N satisfies condition 1 of theorem 10, hence condition 1 of theorem 11, hence is $\mathbb{Z}[G]$ projective. We have $1 \Rightarrow 3$. □

Theorem 13. Let A, B be G-modules, with A cohomologially trivial. In order for $A \otimes B$ to be cohomologically trivial (respectively $\mathrm{Hom}(A, B)$, $\mathrm{Hom}(B, A)$, it is necessary and sufficient that $\mathrm{Tor(A, B)}$ (respectively $\mathrm{Ext}(A, B)$, $\mathrm{Ext}(B, A)$ be.

Proof. By theorem 12, there is an exact sequence:

$$0 \longrightarrow P_1 \longrightarrow P_0 \longrightarrow A \longrightarrow 0$$

with P_i projective. Then we have the exact sequences:

$$0 \longrightarrow \mathrm{Tor}(A, B) \longrightarrow P_1 \otimes B \longrightarrow P_0 \otimes B \longrightarrow A \otimes B \longrightarrow 0$$

$$0 \longrightarrow \mathrm{Ext}(A,B) \longrightarrow \mathrm{Hom}(P_1,B) \longrightarrow \mathrm{Hom}(P_0,B) \longrightarrow \mathrm{Hom}(A,B) \longrightarrow 0$$

By proposition 15, $P_i \otimes B$ and $\mathrm{Hom}(P_i,B)$ are cohomologically trivial so lemma 6 shows that $A \otimes B$, $\mathrm{Hom}(A,B)$ are cohomologically trivial if and only if $\mathrm{Tor}(A,B)$, $\mathrm{Ext}(A,B)$ are respectively. For $\mathrm{Ext}(B,A)$, since A is cohomologially trivial, the induced long exact sequence gives the following short exact sequence:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Hom}(B,P_1) & \longrightarrow & \mathrm{Hom}(B,P_0) & \longrightarrow & \mathrm{Hom}(B,A) & & \\ & & & \searrow & & & & & \\ & & & & \mathrm{Ext}(B,P_1) & \longrightarrow & \mathrm{Ext}(B,P_0) & \longrightarrow & \mathrm{Ext}(B,A) \longrightarrow 0 \end{array}$$

and we conclude in the exact same way. □

Theorem 14. Let G be a finite group and A, A' G-modules. Let $f : A' \to A$ be a G-module homomorphism. For each prime p, let G_p be a Sylow p-subgroup. Suppose for each p $\exists n_p$ such that:

$$\hat{\mathrm{H}}^q(G_p,A') \xrightarrow{f_q^*} \hat{\mathrm{H}}^q(G_p,A)$$

is surjective for $q = n_p$, bijective for $q = n_p + 1$ and injective for $q = n_p + 2$. If B is a G module such that $\mathrm{Tor}(A,B) = \mathrm{Tor}(A',B) = 0$ then:

$$\hat{\mathrm{H}}^q(g, A' \otimes B) \longrightarrow \hat{\mathrm{H}}^q(g, A \otimes B)$$

is an isomorphism for all subgroups g of G and for all $q \in \mathbb{Z}$.

Proof. Let $\bar{A}'$ the an coinduced module $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A')$ so that $A' \hookrightarrow \bar{A}'$. Note that since G is finite, $\bar{A}' \cong \mathbb{Z}[G] \otimes_{\mathbb{Z}} A'$. We see then that $\bar{A}'$ is induced and A' is a direct factor of $\bar{A}'$ as a $\mathbb{Z}$-module. Let $A^* = A \oplus \bar{A}'$. Then A' injects into A^*. If A'' is the cokernel of this injection, we have the following exact sequence:

$$0 \longrightarrow A' \longrightarrow A^* \longrightarrow A'' \longrightarrow 0$$

The long exact sequence induced from this along with the fact that $\bar{A}'$ is cohomologically trivial (induced) gives the long exact sequence:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & \hat{\mathrm{H}}^n(G_p,A') & \longrightarrow & \hat{\mathrm{H}}^n(G_p,A) & \longrightarrow & \hat{\mathrm{H}}^n(G_p,A'') & & \\ & & & \searrow & & & & & \\ & & & & \hat{\mathrm{H}}^{n+1}(G_p,A') & \longrightarrow & \hat{\mathrm{H}}^{n+1}(G_p,A) & \longrightarrow & \hat{\mathrm{H}}^{n+1}(G_p,A'') \longrightarrow \cdots \end{array}$$

The assumptions on n_p show

$$\ker(\hat{\mathrm{H}}^{n_p}(G_p,A) \to \hat{\mathrm{H}}^{n_p}(G_p,A'')) = \ker(\hat{\mathrm{H}}^{n_p+1}(G_p,A) \to \hat{\mathrm{H}}^{n_p+1}(G_p,A'')) = 0$$
$$\mathrm{im}(\hat{\mathrm{H}}^{n_p}(G_p,A'') \to \hat{\mathrm{H}}^{n_p+1}(G_p,A')) = \mathrm{im}(\hat{\mathrm{H}}^{n_p+1}(G_p,A'') \to \hat{\mathrm{H}}^{n_p+2}(G_p,A')) = 0$$

showing $\hat{\mathrm{H}}^{n_p}(G_p,A'') = \hat{\mathrm{H}}^{n_p+1}(G_p,A'') = 0$. By theorem 12, A'' is cohomologically trivial. Now, A^* is a direct sum of A and some copies of A'. The assumptions on B show $\mathrm{Tor}(A^*,B) = 0$,

thus $\mathrm{Tor}(A'', B) = 0$ as A'' is the cokernel of the injection $A' \hookrightarrow A^*$. By theorem 13, $A'' \otimes_{\mathbb{Z}} B$ is cohomologically trivial. Now, since $\mathrm{Tor}(A'', B) = 0$, the following sequence is exact:

$$0 \longrightarrow A' \otimes B \longrightarrow A^* \otimes B \longrightarrow A'' \otimes B \longrightarrow 0$$

The long exact sequence induced by this, along with the fact that $A'' \otimes_{\mathbb{Z}} B$ is cohomologically trivial gives:

$$0 \longrightarrow \hat{\mathrm{H}}^n(g, A' \otimes B) \longrightarrow \hat{\mathrm{H}}^n(g, A^* \otimes B) \longrightarrow 0$$

However, since $A^* \cong A \oplus \bar{A}'$ and $\bar{A}'$ is induced (hence cohomologically trivial) we get:

$$0 \longrightarrow \hat{\mathrm{H}}^n(g, A' \otimes B) \longrightarrow \hat{\mathrm{H}}^n(g, A \otimes B) \longrightarrow 0$$

which gives the result. □

Definition 2.3.2.2 (Cup Product)**.** If G is a finite group with A, B G-modules then the cup product is the unique [4] family of homomorphisms:

$$\hat{\mathrm{H}}^p(G, A) \otimes \hat{\mathrm{H}}^q(G, B) \to \hat{\mathrm{H}}^{p+q}(G, A \otimes B)$$

Theorem 15. Again, let G be a finite group with A, B, C G-modules. Let $\varphi : A \times B \to C$ be a G-invariant map. Let $a \in \hat{\mathrm{H}}^q(G, A)$. For any subgroup $g \leq G$ and any G-module D we write $f(n, g, D) : \hat{\mathrm{H}}^n(g, B \otimes D) \to \hat{\mathrm{H}}^n(g, C \otimes D)$ given by the cup product with $a_g := \mathrm{Res}_{G/g}(a)$. Now suppose for each prime p and every Sylow-p subgroup G_p there exists n_p such that $f(n, G_p, \mathbb{Z})$ is surjective for $n = n_p$, bijective for $n = n_p + 1$ and injective for $n = n_p + 2$. Then $f(n, g, D)$ is bijective for all n, g, D such that $\mathrm{Tor}(B, D) = \mathrm{Tor}(C, D) = 0$.

Proof. First we look at the case $q = 0$. $a \in \hat{\mathrm{H}}^0(G, A)$ is represented by $a' \in A^G$. Letting $f(b) = \varphi(a, b)$ we get a G-homomorphism from B to C. Then the cup product morphism $f(n, g, D)$ is the morphism induced by $f \otimes 1 : B \otimes D \to C \otimes D$. This puts us in the realm of theorem 14 and gives us the result for $q = 0$. We move now to the general case. Similarly to the last theorem, let $\bar{A}$ be the natural induced module such that $A \hookrightarrow \bar{A}$. Let $A_1 = \bar{A}/A$. Let $\bar{C}, C_1$ be defined analogously for C and $\varphi_1 : A_1 \times B \to C_1$. Then $a \in \hat{\mathrm{H}}^q(G, A)$ can be written $\delta(a_1)$ for $a_1 \in \hat{\mathrm{H}}^{q-1}(G, A_1)$ where δ is the connecting homomorphism in the long exact sequence induced by:

$$0 \longrightarrow A \longrightarrow \bar{A} \longrightarrow A_1 \longrightarrow 0$$

Note that δ is an isomorphism since $\bar{A}$ is cohomologically trivial. Using the cup product with a_1 and φ_1 we get homomorphisms $f_1(n, g, D) : \hat{\mathrm{H}}^n(g, B \otimes D) \to \hat{\mathrm{H}}^n(g, C_1 \otimes D)$. Composing with the connecting homomorphism (isomorphism since $\bar{C}$ is cohomologically trivial) from the long exact sequence induced by:

$$0 \longrightarrow C \longrightarrow \bar{C} \longrightarrow C_1 \longrightarrow 0$$

We get $f(n, g, D)$. Using the fact that cup products commute with coboundaries [1] we see if it holds for a_1 it holds for a. Hence we can pass from $q - 1$ to q. To pass from q to $q - 1$ we let A_2 be a free module such that $A = A_2/Q$ and similarly for C_2 and φ_2. Then the fact that A_2 is cohomologically trivial shows we can write the class $a \in \hat{\mathrm{H}}^{q-1}(G, A)$ can be written $\delta^{-1}(a_1)$ for $a_1 \in \hat{\mathrm{H}}^q(G, A_1)$ where δ is the connecting homomorphism. Then then we have similar homomorphisms $f_2(n, g, D) : \hat{\mathrm{H}}^n(g, B \otimes D) \to \hat{\mathrm{H}}^n(g, C_1 \otimes D)$ and conclude in the same way. Hence we can pass from q to $q - 1$. □

We now show some special cases of the previous theorem that might help motivate the abstract jumble we've just gone through.

Theorem 16. Let G be a finite group, A a G-module, and $a \in \mathrm{H}^2(G, A)$. For each prime p, let G_p be a Sylow-p subgroup. Finally suppose that for each G_p:

1. $\mathrm{H}^1(G_p, A) = 0$
2. $\mathrm{H}^2(G_p, A)$ is generated by $\mathrm{Res}_{G/G_p}(a)$ and has order equal to $|G_p|$

Then for any G-module D such that $\mathrm{Tor}(A, D) = 0$, the cup product with $a_g = \mathrm{Res}_{G/G_p}(a)$ induces isomorphisms:

$$\hat{\mathrm{H}}^n(g, D) \longrightarrow \hat{\mathrm{H}}^{n+2}(g, A \otimes D)$$

Proof. Apply theorem 15 with $A = A, B = \mathbb{Z}, C = A$, $q = 2$, $\varphi : A \times \mathbb{Z} \to A$ being the projection onto the first coordinate. Use $n_p = -1$. Condition 1 shows $f(-1, G_p, \mathbb{Z})$ is surjective. Condition 2 shows the bijection. Finally since $\mathbb{Z}$ is a trivial G-module, $\mathrm{H}^1(G_p, \mathbb{Z}) = \mathrm{Hom}(G_p, \mathbb{Z}) = 0$ which shows the injection needed. □

By applying this theorem with $D = \mathbb{Z}$ we get:

Corollary 3 (Tate)**.** For every $n \in \mathbb{Z}$ and every subgroup $g \leq G$, the cup product with $a_g = \mathrm{Res}_{G/g}(a)$ defines an isomorphism $\hat{\mathrm{H}}^n(g, \mathbb{Z}) \to \hat{\mathrm{H}}^{n+2}(g, A)$.

Chapter 3

Main Results

We now look to apply some of the background knowledge to understand the local field extensions.

3.1 Galois Cohomology

The first thing we have to develop is Galois cohomology. We have build a solid understanding of certain topics in group cohomology already and will use this to understand local fields as Galois group modules. Let $G = \mathrm{Gal}(L/K)$ be the Galois group of a finite extension. G acts on both the additive group L and the multiplicative group L^* via automorphisms so we can view these as G-modules.

Proposition 16. For all $n \in \mathbb{Z}$ $\hat{\mathrm{H}}^n(G, K) = 0$

Proof. (Weibel [4]) Normal basis theorem [6] shows that K is an induced G module, hence cohomologically trivial. □

This is not particularly interesting and does not tell us much. Therefore, we will concern ourselves with the multiplicative group.

3.2 Brauer Group

The Brauer group of a field is often defined via central simple algebras [7], but it can also be described cohomologically.

Proposition 17. Let K/F be a Galois extension with Galois group G. Then $\mathrm{H}^1(G, K^*) = 0$

Proof. (Serre [1]) Let $\varphi : s \mapsto a_s$ be a 1-cocycle. For $c \in K^*$ we can form the series:

$$b = \sum_{s \in G} a_s s(c)$$

Since $s \in G$ is an automorphism on K and automorphisms are linearly independent, we can

choose c such that $b \neq 0$. Now:

$$\begin{aligned} s(b) &= \sum_{t \in G} s(a_t) \cdot st(c) \\ &= \sum_{t \in G} a_s^{-1} a_{st} \cdot st(c) \\ &= a_s^{-1} \cdot b \end{aligned}$$

where the second equality comes from:

$$\begin{aligned} & a_t = \varphi(t) \\ \Rightarrow & s(a_t) = s \cdot \varphi(t) \\ = & \varphi(st) - \varphi(s) \text{ (using the fact that } \varphi \text{ is a 1-cocycle)} \\ = & a_{st} \cdot a_s^{-1} \text{ (using the fact that group action moves to multiplication in } K^*) \end{aligned}$$

Now we see that $s(b) = a_s^{-1} \cdot b = -\varphi(s) + b$ which shows $\varphi(s) = b - s(b)$ and hence φ is a coboundary as well. □

One corollary of this is the famous Hilbert's theorem 90 [8]. For K/F a Galois extension with Galois group G, we write $\mathrm{H}^n(G, K^*) = \mathrm{H}^n(K/F)$ for brevity since the cohomology groups depend only on the field extension. We now can use the previous proposition to look at an inflation restriction sequence:

Proposition 18. Let L/F be a Galois extension containing the Galois extension L/K. Then there is an exact sequence:

$$0 \longrightarrow \mathrm{H}^n(K/F) \longrightarrow \mathrm{H}^n(L/F) \longrightarrow \mathrm{H}^n(L/K)$$

Proof. Note that if $G = \mathrm{Gal}(L/F), H = \mathrm{Gal}(L/K)$. Then H is normal in G and $\mathrm{Gal}(K/F) = G/H$. Additionally, proposition 17 shows $\mathrm{H}^1(H, L^*) = 0$. We can then apply proposition 12 to get the exact sequence:

$$0 \longrightarrow \mathrm{H}^2(G/H, K^*) \longrightarrow \mathrm{H}^2(G, L^*) \longrightarrow \mathrm{H}^2(H, L^*)$$

giving the result. □

We can pass to the limit on L to get the exact sequence:

$$0 \longrightarrow \mathrm{H}^2(K/F) \longrightarrow \mathrm{H}^2(F_s/F) \longrightarrow \mathrm{H}^2(K_s/K)$$

where F_s and K_s are the separable closures of F and K respectively.

Definition 3.2.0.1. Let F be a local field and F_s the separable closure of F. Then we call $\mathrm{H}^2(F_s/F)$ the Brauer group of F and denote it B_F.

Definition 3.2.0.2. We say an element $a \in B_F$ is split by K if it is in the kernel of the map $B_F \to B_K$.

We show some examples below (worked out by the author).

Example 5. We compute the Brauer group of $\mathbb{R}$. We know that the separable (algebraic) closure of $\mathbb{R}$ is $\mathbb{C}$ and this is a degree 2 extension having Galois group $\mathbb{Z}/2\mathbb{Z}$. Hence we look to compute $\mathrm{H}^2(\mathbb{C}/\mathbb{R}) = \mathrm{H}^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^*)$. Cyclic group cohomology has well defined formulas [4] so this is quite easy. We get:

$$\begin{aligned} B_{\mathbb{R}} &= \mathrm{H}^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^*) \\ &= \mathbb{C}^{*G}/N\mathbb{C}^* \\ &= \mathbb{R}^*/\mathbb{R}_{>0} \end{aligned}$$

Where the last line follows since $N\mathbb{C}^*$ is just $N = 1 + (\bar{-})$ applied to the multiplicative group $\mathbb{C}^*$ giving the set $\{z \cdot \bar{z} : z \in \mathbb{C}\} = ||\mathbb{C}||^2$. Finally we conclude $\mathbb{R}^*/\mathbb{R}_{>0} = \mathbb{Z}/2\mathbb{Z}$. This coincides with the classical computation, where the nontrivial element corresponds to the quaternion algebra [7].

Example 6. Let $\mathbb{F}_q$ be a finite field. I claim the Brauer group of a $\mathbb{F}_q$ is trivial. We first note that every Galois extension of $\mathbb{F}_q$ is cyclic and we can find cyclic extensions of degree n for any n by using $\mathbb{F}_{q^n}$. Hence, by passing to the limit, we get that the absolute Galois group is $\hat{\mathbb{Z}}$. For a finite subextension we have (using well known cyclic group cohomology formula) $\mathrm{H}^2(\mathbb{Z}/n\mathbb{Z}, \mathbb{F}_{q^n}^*) = (\mathbb{F}_{q^n}^*)^{\mathbb{Z}/n\mathbb{Z}}/N\mathbb{F}_{q^n}^*$. However, the norm map on $\mathbb{F}_{q^n}^*$ is surjective. Hence this cohomology group is 0. Now:

$$\begin{aligned} \mathrm{H}^2(\varprojlim \mathbb{Z}/n\mathbb{Z}, \mathbb{F}_{q^n}^*) &= \varinjlim \mathrm{H}^2(\hat{\mathbb{Z}}/n\hat{\mathbb{Z}}, (\mathbb{F}_{q^n}^*)^{n\hat{\mathbb{Z}}}) \\ &= \varinjlim \mathrm{H}^2(\mathbb{Z}/n\mathbb{Z}, (\mathbb{F}_{q^n}^*)^{n\hat{\mathbb{Z}}}) \\ &= \varinjlim \mathrm{H}^2(\mathbb{Z}/n\mathbb{Z}, \mathbb{F}_{q^n}^*) \\ &= 0 \end{aligned}$$

Finally, passing to the limit on the $\mathbb{F}_{q^n}^*$ we get the result.

Example 7. The Brauer group of $\mathbb{Q}_p$ is isomorphic to $\mathbb{Q}/\mathbb{Z}$. This requires a bit more work but we will come back to this to show it is true! [9]

From here on in this section we will have K be a field complete under a discrete valuation with a perfect residue field $\bar{K}$.

Proposition 19. Let K_{nr} be the maximal unramified extension of K. Then $\mathrm{H}^q(K_s/K) = \mathrm{H}^q(K_{nr}/K)$.

We will use the following lemma which can be found in chapter 8 of [10].

Lemma 7. $B_{K_{nr}} = 0$

The following proof was done by the author with the help of some notes on spectral sequences (Gregory [11]).

Proof. Let L/K be an unramified extension. We have the short exact sequence:

$$1 \longrightarrow \mathrm{Gal}(K_s/L) \longrightarrow \mathrm{Gal}(K_s/K) \longrightarrow \mathrm{Gal}(L/K) \longrightarrow 1$$

We have a directed system of cohomology with inflation homomorphisms, so we can pass to the limit and get:

$$1 \longrightarrow \mathrm{Gal}(K_s/K_{nr}) \longrightarrow \mathrm{Gal}(K_s/K) \longrightarrow \mathrm{Gal}(K_{nr}/K) \longrightarrow 1$$

Thus $\mathrm{Gal}(K_{nr}/K) = \mathrm{Gal}(K_s/K)/\mathrm{Gal}(K_s/K_{nr})$. Now we want to use the Lyndon-Hochschild-Serre spectral sequence [12] [4]. First we calculate $\mathrm{H}^j(\mathrm{Gal}(K_s/K_{nr}), K_s^*)$. Proposition 17 shows that
$\mathrm{H}^1(\mathrm{Gal}(K_s/K_{nr}), K_s^*) = 0$. The lemma above shows $\mathrm{H}^2(\mathrm{Gal}(K_s/K_{nr}), K_s^*) = B_{K_{nr}} = 0$. Then theorem 10 tells us that K_s^* is cohomologically trivial as a $\mathrm{Gal}(K_s/K_{nr})$-module. Hence $\mathrm{H}^j(\mathrm{Gal}(K_s/K_{nr}), K_s^*) = 0$ for $j > 0$ (where cohomology and Tate cohomology coincide). Looking at the E_2 page of our spectral sequence we see $E_2^{i,j} = 0$ for $j > 0$. We see the E_2 page collapses to the $E_2^{i,0}$ and thus maps in and out of these must be trivial. Hence $E_2^{i,j} = E_\infty^{i,j}$ for all i, j [11]. Since j is always 0 in our case, we simply read off:

$$\begin{aligned}
&\mathrm{H}^i(\mathrm{Gal}(K_s/K), K_s^*) \\
=&\mathrm{H}^{i+0}(\mathrm{Gal}(K_s/K), K_s^*) \\
=&E_2^{i,0} \\
=&\mathrm{H}^i(\mathrm{Gal}(K_{nr}/K), \mathrm{H}^0(\mathrm{Gal}(K_s/K_{nr}), K_s^*)) \\
=&\mathrm{H}^i(\mathrm{Gal}(K_{nr}/K), (K_s^*)^{\mathrm{Gal}(K_s/K_{nr})}) \\
=&\mathrm{H}^i(\mathrm{Gal}(K_{nr}/K), K_{nr}^*)
\end{aligned}$$

□

Corollary 4. $B_K = \mathrm{H}^2(K_{nr}/K)$

Corollary 5. Every element of B_K is split by some finite unramified extension of K.

Proof. $B_{K_{nr}} = 0$, hence the image of $a \in B_K$ in $B_{K_{nr}}$ is 0. Also, $B_K = \mathrm{H}^2(K_{nr}/K) = \varinjlim \mathrm{H}^2(L/K)$ so the image of a is zero in one of the B_L. □

Corollary 6. B_K is the union of the $\mathrm{H}^2(L/K)$ as L runs over the finite unramified extensions of K.

Thus we simply need to determine the $\mathrm{H}^2(L/K)$. Since L/K is unramified, we can identity L^* with $U_L \times \mathbb{Z}$ [10]. Thus we get a split exact sequence of cohomology:

$$0 \longrightarrow \mathrm{H}^q(G, U_L) \longrightarrow \mathrm{H}^q(L/K) \longrightarrow \mathrm{H}^q(G, \mathbb{Z}) \longrightarrow 0$$

Lemma 8. For all $q \geq 1$ $\mathrm{H}^q(G, U_L^1) = 0$

This is proven by Serre [1] using some cochain and cocycle magic. This and the exact sequence which follows from proposition 5:

$$0 \longrightarrow U_L^{(1)} \longrightarrow U_L \longrightarrow \bar{L}^* \longrightarrow 0$$

shows that $\mathrm{H}^q(G, U_L) = \mathrm{H}^q(G, \bar{L}^*)$. Since $G = \mathrm{Gal}(L/K) = \mathrm{Gal}(\bar{L}/\bar{K})$, we get $\mathrm{H}^q(G, U_L) = \mathrm{H}^q(\bar{L}/\bar{K})$. Then replacing $\mathrm{H}^q(G, U_L)$ in the first exact sequence:

$$0 \longrightarrow \mathrm{H}^q(\bar{L}/\bar{K}) \longrightarrow \mathrm{H}^q(L/K) \longrightarrow \mathrm{H}^q(G, \mathbb{Z}) \longrightarrow 0$$

$$0 \longrightarrow \mathrm{H}^q(\bar{K}_{nr}) \longrightarrow \mathrm{H}^q(K_{nr}/K) \longrightarrow \mathrm{H}^q(G, \mathbb{Z}) \longrightarrow 0$$

Now consider the exact sequence:

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

$\mathbb{Q}$ is an injective module and thus has trivial cohomology, giving us the isomorphism:

$$0 \longrightarrow \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \longrightarrow \mathrm{H}^2(G, \mathbb{Z}) \longrightarrow 0$$

Lemma 9. Let K be a local field. Then $G = \mathrm{Gal}(K_{nr}/K) = \hat{\mathbb{Z}}$

Proof. (Serre [1]) If L/K is unramified then its Galois group is isomorphic to $\mathrm{Gal}(\bar{L}/\bar{K})$. This is a finite extension and thus is cyclic, generated by the Frobenius automorphism. It is of order $n = [\bar{L} : \bar{K}]$ Then $\mathrm{Gal}(\bar{L}/\bar{K}) = \mathrm{Gal}(L/K) = \mathbb{Z}/n\mathbb{Z}$. Passing to the projective limit we get the result. □

We just need to determine $\mathrm{Hom}(\hat{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z})$.

Proposition 20. $\mathrm{Hom}(\hat{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}) = \mathbb{Q}/\mathbb{Z}$

Proof. The following proof is done by the author using some category theoretic facts about limits and colimits [11].

$$\begin{aligned}
&\mathrm{Hom}(\hat{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}) \\
=&\mathrm{Hom}(\varprojlim \mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \\
=&\varinjlim \mathrm{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \text{ (using contravariance of Hom in the first argument)} \\
=&\varinjlim \mathbb{Z}/n\mathbb{Z} \\
=&\mathbb{Q}/\mathbb{Z}
\end{aligned}$$

□

We then get the exact sequence:

$$0 \longrightarrow B_{\bar{K}} \longrightarrow B_K \xrightarrow{\mathrm{inv}_K} \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

We are now ready to compute example 7. By example 6, $B_{\mathbb{F}_p} = 0$. Therefore the above exact sequence becomes

$$0 \longrightarrow B_{\mathbb{Q}_p} \xrightarrow{\mathrm{inv}_{\mathbb{Q}_p}} \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

giving us the isomorphism needed. In fact one can see this holds for any local field with finite residue field. We now quote some results from Harari [10] about the properties of $\mathrm{inv_K}$.

Proposition 21. Let L/K be a finite extension of degree n. Then the following diagram commutes:

$$\begin{array}{ccc}
B_K & \xrightarrow{\mathrm{inv_K}} & \mathbb{Q}/\mathbb{Z} \\
{\scriptstyle \mathrm{Res_{K/L}}}\downarrow & & \downarrow{\scriptstyle n} \\
B_L & \xrightarrow[\mathrm{inv_L}]{} & \mathbb{Q}/\mathbb{Z}
\end{array}$$

Corollary 7. An element $a \in B_K$ is split by $L \Leftrightarrow na = 0$

This follows from the diagram above.

Corollary 8. Let L/K be Galois. Then the isomorphism $\mathrm{inv_K}$ maps the subgroup $\mathrm{H}^2(L/K)$ of B_K to the order n subgroup of $\mathbb{Q}/\mathbb{Z}$.

Proof. Serre simply says this is clear, but one might desire a bit more detail. We have that if $b \in \mathrm{H}^2(L/K) \Leftrightarrow$ it's image in B_L is 0 as seen by the union characterization of the Brauer group. In particular, by the corollary above, $nb = 0$. Thus $\mathrm{H}^2(L/K)$ maps to a subgroup of $\mathbb{Q}/\mathbb{Z}$ with order dividing n. However since inv_K is an isomorphism, we can consider $a := \mathrm{inv}_K^{-1}(1/n) \in B_K$. Clearly $na = \mathrm{inv_K}^{-1}(0) = 0$ hence $a \in \mathrm{H}^2(L/K)$. □

3.3 Class Formations and the Local Artin Map

We now look at the setting in which we will apply the powerful group cohomology facts from the background chapter.

Definition 3.3.0.1 (Formation)**.** A formation $(G, \{G_E\}_{E\in X}, A)$ is a group G, a family of subgroups G_E for E in X an index set, and a G-module A satisfying the following properties:

1. $G_E = G_{E'} \Rightarrow E = E'$
2. for every finite collection $F_i \in X$, $\exists F \in X$ such that $G_F = \cap_i G_{F_i}$
3. every subgroup G' of G which contains some G_F, $F \in X$, has the form $G' = G_{F'}$ for some $F' \in X$
4. if $s \in G$ and $F \in X$, $sG_Fs^{-1} = G_{F''}$ for some $F'' \in X$.
5. for all $a \in A$, $G \geq \mathrm{stab}(a) = G_E$ for some $E \in X$

We denote $A^{G_E} = A_E$ and condition 5 above ensures that $A = \cup_{E\in X} A_E$. This concept we will apply to Galois extensions! We take F/E to be Galois $G = \mathrm{Gal}(F/E)$, and X to be the set of subfields of F which are extensions of E. Then we see that $G_{F'} = \mathrm{Gal}(F/F')$. Often we take F to be the separable closure of E. We can then study the Galois cohomology of these formations which leads us into our next definition:

Definition 3.3.0.2 (Class Formation)**.** A class formation is a formation $(G, \{G_E\}_{E\in X}, A)$ along with a homomorphism $\mathrm{inv_E} : \mathrm{H}^2(E_s/E) \to \mathbb{Q}/\mathbb{Z}$ (where E_s is again the separable closure of E) satisfying the following properties:

1. For each Galois extension F/E, $\mathrm{H}^1(F/E) = 0$
2. inv_E is injective and maps $\mathrm{H}^2(F/E)$ to the unique order $[F : E]$ subgroup of $\mathbb{Q}/\mathbb{Z}$.
3. For any extension E'/E we have $\mathrm{inv}_{E'} \circ \mathrm{Res}_{E'/E} = [E' : E] \cdot \mathrm{inv}_E$

We note that we have a directed system of cohomology via inflation homomorphisms and can think of $\mathrm{H}^q(E_s/E)$ as the direct limit of $\mathrm{H}^q(F/E)$ for F/E Galois. This shows $\mathrm{H}^q(E_s/E) = \cup_F \mathrm{H}^q(F/E)$. Let us now verify that our Galois formation is indeed a class formation. Let E be a local field and E_s the separable closure. Via some Galois theory, one can see this is indeed a formation with X being the finite subextensions. We have seen in prop 17 that for every $F \in X$, $\mathrm{H}^1(F/E) = 0$. Condition 2 is satisfied by the second corollary to proposition 21. Condition 3 is proposition 21 itself. We now look to apply Tate's theorem, corollary to theorem 16. Let $F \in X$. For any F' a subextension of F, proposition 17 shows that $\mathrm{H}^1(F/F') = 0$. In particular, any Sylow-p subgroup corresponds to some subextension $\mathrm{Gal}(F/F')$. Thus condition 1 is satisfied. Now for the same subextension we have that $\mathrm{H}^2(F/F')$ is cyclic of order $\mathrm{Gal}(F/F')$. We therefore have some $a_{F/F'} \in \mathrm{H}^2(F/F')$ which generates the group. Hence 2 is satisfied. We can then apply Tate's to say:

Theorem 17. For all $n \in \mathbb{Z}$, the cup product with $a_{F/E}$ defines an isomorphism:

$$\hat{\mathrm{H}}^n(\mathrm{Gal}(F/E), \mathbb{Z}) \longrightarrow \hat{\mathrm{H}}^{n+2}(\mathrm{Gal}(F/E), F^*)$$

We look specifically at the $n = -2$ case.

Corollary 9. There is an isomorphism:

$$\mathrm{Gal}(F/E)^{ab} \longrightarrow E^*/NF^*$$

Proof. We have seen in example 4 that $\hat{\mathrm{H}}^{-2}(\mathrm{Gal}(F/E),\mathbb{Z}) = \mathrm{H}_1(\mathrm{Gal}(F/E),\mathbb{Z}) = G^{ab}$. Now by definition we have $\hat{\mathrm{H}}^{0}(\mathrm{Gal}(F/E),F^*) = (F^*)^{\mathrm{Gal}(F/E)}/NF^* = E^*/NF^*$. □

The inverse to this isomorphism is called the local Artin Reciprocity isomorphism or the local Artin map. It is denoted θ and has some important functorial properties which can be found in Serre [1], chapter 11 section 3. These properties allow us to pass to the limit and deduce isomorphisms for infinite extensions.

3.4 Existence Theorem

We have shown a nice isomorphism between the abelianization of a Galois group and the norm quotient groups. We now look to show that for any norm group, we have a corresponding abelian Galois extension. This is called the existence theorem.

Definition 3.4.0.1 (Norm map in arbitrary extension)**.** Let E'/E be an arbitrary field extension. Take s_i be the coset representatives of $\mathrm{Gal}(F/E)/\mathrm{Gal}(F/E')$. Then for $a' \in E'^*$ the norm map $N_{E'/E}$ sends a' to $\prod_i s_i a'$.

This does not depend on coset reps since the classes of the quotient group all act on E' in the same way. It indeed maps to E^* since, as seen in the background chapter, the norm applied to E'^* is in the fix of G. When the extension is Galois we get our usual norm map.

Definition 3.4.0.2 (Norm Group)**.** A norm group $I \leq E^*$ is a subgroup such that there exists E'/E with $N_{E'/E}(E'^*) = I$.

Proposition 22. Let E'/E be an arbitrary extension with E'' the largest abelian subextension. Then $N_{E'/E}(E'^*) = N_{E''/E}(E''^*)$.

We will use the following lemma which is one of the functorial properties of the reciprocity map found in Serre [1]. We will also use the transitivity of the norm map.

Lemma 10. The following diagram commutes:

$$\begin{array}{ccc} E'^* & \xrightarrow{Norm} & E^* \\ \downarrow & & \downarrow \\ \mathrm{Gal(F/E')}^{ab} & \longrightarrow & \mathrm{Gal(F/E)}^{ab} \end{array}$$

Proof. (Serre [1]) Let F/E be a Galois extension containing E'. Let G' be the commutator subgroup of G. Let $H = \mathrm{Gal}(F/E')$. Then $\mathrm{Gal}(F/E'') = G' \cdot H$. Let $a \in N_{E''/E}(E''^*)$. Then it is the identity in E^*/NE''^* and it's image under the reciprocity isomorphism (denoted $(a, E''/E)$) is also the identity in the quotient group $G/(G' \cdot H)$. If we take the image of a under the reciprocity map for F/E we get the element $(a, F/E) \in G^{ab} = G/G'$. Since $(a, F/E)$ is the identity in $G/(G'H)$, it must lie in the image of $H/H' \hookrightarrow G/G'$. Now since $E'^*/NF^* \cong H/H'$ we get the surjection $E'^* \to H/H'$. This and the above commutative diagram show that $\exists a' \in E'^*$ such that $(a, F/E) = (N_{E'/E}(a'), F/E)$. Therefore $(a, F/E) - (N_{E'/E}(a'), F/E) = 0 \in$

$\mathrm{Gal}(F/E)^{ab}$ implies that $a - N_{E'/E}(a') \in N_{F/E}(F^*)$ (as it is the identity E^*/NF^*). Thus $\exists a'' \in F^*$ such that $N_{F/E}(a'') = a - N_{E'/E}(a')$. Thus $a = N_{E'/E}(a') + N_{F/E}(a'') = N_{E'/E}(a' + N_{F/E'}(a''))$. We see $a \in N_{E'/E}(E'^*)$. Now let $a \in N_{E'/E}(E'^*)$. Then $a \in N_{E''/E}(N_{E'/E''}(E'^*))$ and $a \in N_{E''/E}(E''^*)$. Two way inclusion is shown □

Corollary 10. $N_{E'/E}(E^*)$ has finite index in E^*. It divides $[E' : E]$ and is equal iff E'/E is abelian.

Proof. We have $[E^* : N_{E'/E}(E'^*)] = [E^* : N_{E''/E}(E''^*)] = |G^{ab}| = [E'' : E]$ which is finite and divides $[E' : E]$ with equality holding if E'/E is abelian. □

To simplify notation and for ease of writing set $I_F := N_{F/E}(F^*)$.

Proposition 23. The map $F \to I_F$ is a bijection between the set of abelian extensions of E and the set of norm groups of E^*. This reverses inclusions and satisfies:

1. $I_{F \cdot F'} = I_F \cap I'_F$
2. $I_{F \cap F'} = I_F + I_{F'}$

In addition, any subgroup of E^* containing a norm group is itself a norm group.

Proof. (Serre [1]) If F, F' are abelian extensions of E, then so is $F \cdot F'$. Also by transitivity of norms (similar to above), we see $I_{F \cdot F'} \subset I_F \cap I_{F'}$. So let $a \in I_F \cap I_{F'}$. The element $(a, (F \cdot F')/E)$ has trivial image in both $\mathrm{Gal}(F/E)$ and $\mathrm{Gal}(F'/E)$ which shows it is indeed trivial in $\mathrm{Gal}((F \cdot F')/E)$ and thus is in $I_{F \cdot F'}$. In the case $I_F \subset I_{F'}$ then $I_{F \cdot F'} = I_F \cap I_{F'} = I_F$ and thus $[F \cdot F' : E] = [F : E]$ thus $F \supset F'$. So we indeed have that this is bijective and reverses inclusions. The rest of the proof was considered immediate but the author did not feel the same way. The above shows that $I_{F \cap F'} \supset I_F + I_{F'}$. Let K be the field that gets mapped to $I_F + I_{F'}$. Then $K \supset F \cap F'$ and $I_{F \cap F'} = I_{(F \cap F') \cdot K} = I_{F \cap F'} \cap (I_F + I_{F'}) = I_F + I_{F'}$ by the inclusion above. Finally, let $I \supset I_F$ for I_F a norm group. Then we have:

$$\begin{aligned} & I_F \leq I \leq E^* \\ \Rightarrow & I/I_F \leq E^*/I_F = \mathrm{Gal}(F/E)^{ab} \\ \Rightarrow & I/I_F \cong \mathrm{Gal}(F/E')^{ab} \cong E'^*/I_F \end{aligned}$$

We then see that I/I_F is trivial in $\mathrm{Gal}(F/E)^{ab}/\mathrm{Gal}(F/E')^{ab} = \mathrm{Gal}(E'/E)^{ab} \cong E^*/NE'^*$ so this quotient is a norm group. Then $I = I/I_F + I_F$ which by above is also a norm group, showing the final claim. □

Definition 3.4.0.3. The universal norm group of E, D_E, is the intersection of all norm groups of E^*.

We work with the multiplicative group of local fields as a topological group, inheriting the topology from their field. From now on, we suppose:

1. For every extension F/E the norm homomorphism has closed image and compact kernel
2. For each prime number p, there exists a field E_p such that for $F \supset E_p$ $\varphi_p : E^* \to E^*, x \mapsto px$ has compact kernel and image containing D_E

3. There exists a compact subgroup U_E of E^* such that every closed subgroup of finite index in E^* which contains U_E is a norm group.

The next two propositions are preliminaries to the existence theorem. These and the existence theorem itself follow Serre [1].

Proposition 24. For every extension F/E, $N_{F/E}D_F = D_E$

Proof. Let $a \in N_{F/E}D_F$. Then $a \in N_{F/E}(N_{F'/F}F'^*)$ for all F'/F. Thus for all F'/E, $a \in N_{F'/E}(F'^*) \Rightarrow N_{F/E}D_F \subset D_E$. Now let $a \in D_E$ and F'/F be an extension. Define $K(F') := \{b \in F^* : N_{F/E}(b) = a \text{ and } b \in N_{F'/F}(F'^*)\}$. Then $K(F') = N_{F'/F}(F'^*) \cap N^{-1}_{F/E}(a)$. Now, $N^{-1}_{F/E}(a) = \{c \in F : N_{F/E}(c - a) = 0\}$ which is compact by assumption 1. Since $a \in D_E$, there is a $b \in F^*$ such that $N_{F/E}(b') = a$ and the image of b' in F^* is in $N_{F'/F}(F'^*)$ (by universal norm group properties). Thus $K(F')$ is non empty. Then varying F' makes $K(F')$ a directed decreasing family of compact subsets of $K(F)$. We have then that the intersection is non-empty. Take b to be in the intersection. Then by construction, $b \in D_F$ and $N_{F/E}(b) = a$. So one sees that $\forall a \in D_E$, $\exists b \in D_F$ such that $N_{F/E}(b) = a$ implying $N_{F/E}(D_F) \supset D_E$. Two way inclusion is then shown. □

Proposition 25. For every field E, the group D_E is divisible and equal to $\cap_n(n \cdot E^*)$.

Proof. First we show $D_E = pD_E$. Let $a \in D_E$ and let F be an extension of E containing E_p (assumption 2). Define $L(F) := \{b \in E^* : pb = a \text{ and } b \in N_{F/E}(F^*)\}$. This is a compact subset (again assumption 2). By proposition 24, we have that $a = N_{F/E}(x)$ for some $x \in D_F$. Using again assumption 2, we have that $x = py$ for some $y \in F^*$. Then $b = N_{F/E}(y) \in L(F)$ and $L(F) \neq \emptyset$. Analogously to the $K(F)$, the intersection of the $L(F)$ is non-empty. If b lies in this intersection, $pb = a$ and $b \in D_E$. Then since a was arbitrary, we see $D_E = pD_E$ for all p prime. By taking products of these p we get that D_E is divisible. Now we see $D_E \subset \cap_n(n \cdot E^*)$. Let $a \in \cap_n(n \cdot E^*)$. For each extension F/E of degree n, there is some $b \in E^*$ such that $a = nb$ (Note that this language is indicative of the divisibility but is somewhat confusing as in the multiplicative group E^* this means that $a = b^n$).

$$\Rightarrow a = b^n = N_{F/E}(b)$$
$$\Rightarrow a \in N_{F/E}(F^*)$$

Since this is true for any extension, we get that $a \in D_E$. We then get the two way inclusion. □

The following lemmas are basic facts for topological groups but are important to the next theorem.

Lemma 11. Let A be a topological group. Then a subgroup B of finite index is closed iff it is open.

Proof. Let B be closed. Write A as a finite union of cosets of B. Then $B^c = A \setminus B$ is a finite union of closed cosets, hence closed. Thus B is open.
Now let B be open. Again write A as a finite union of cosets of B. Then $B^c = A \setminus B$ is a finite union of open cosets, hence open. Thus B is closed. □

Lemma 12. Let A be a topological group with B a closed subgroup of finite index. If C is another subgroup containing B, then it is also closed and of finite index.

Proof. We immediately see that since $B \leq C$, C has index less than or equal to B. This gives the finiteness of index. Now, we write $C = \cup g_i B$. This is a finite union since B has finite index in A, hence in C. We see C^c is a finite union of cosets of B, so it is closed. Then C is open of finite index and, by the previous lemma 11, is closed of finite index. □

We are now ready to prove the existence theorem. We will restate our assumptions.

Theorem 18 (Existence Theorem)**.** We work with the multiplicative group of local fields as a topological group, inheriting the topology from their field. Suppose:

1. For every extension F/E the norm homomorphism has closed image and compact kernel
2. For each prime number p, there exists a field E_p such that for $F \supset E_p$ $\varphi_p : E^* \to E^*, x \mapsto px$ has compact kernel and image containing D_E
3. There exists a compact subgroup U_E of E^* such that every closed subgroup of finite index in E^* which contains U_E is a norm group.

Then a subgroup of E^* is a norm group iff it is closed of finite index.

Proof. First if $I \leq E^*$ is a norm group, then it is the image of some $N_{F/E}(F^*)$ for a finite extension F/E. By proposition 22, it's index is equal to $[E'' : E]$ where E''/E is the largest abelian subextension of F/E. Thus the index is finite. I is closed by assumption 1.

Now suppose I is a subgroup of E^* which is closed of finite index (thus open of finite index by the lemma 11 above). If $[E^* : I] = n$ then $(E^*)^n \subset I$. Then by proposition 25 $D_E \subset I$. As N runs through the family of norm groups of E^*, we get $\cap(N \cap U_E) = D_E \cap U_E \subset I$. Now each $N \cap U_E$ is compact and I is open. Therefore there exists some N for which $N \cap U_E \subset I$. Let $a \in N \cap (U_E + (N \cap I))$. Then:

$$\begin{aligned}
&\Rightarrow a = a' + a'' \text{ (with } a' \in U_E \cap N \text{ and } a'' \in N \cap I) \\
&\Rightarrow a \in I \\
&\Rightarrow N \cap (U_E + (N \cap I)) \subset I
\end{aligned}$$

$N \cap I$ is the intersection of two groups which are closed of finite index, hence is itself. Then $U_E + (N \cap I)$ is also closed of finite index (by lemma 12) and contains U_E. By assumption 3, it is a norm group. Then the intersection $N \cap (U_E + (N \cap I))$ is as well by proposition 23. Since I contains this group, it is also a norm group (again by proposition 23). □

We now want to check that our usual local field framework indeed satisfies the conditions of the theorem. Let K be a field complete under a discrete valuation with finite residue field, giving K the structure of a locally compact field. Let U_E be the group of units of the ring of integers of E.
Let U be a compact subset of E^*. Then the finite subcover of U is contained in a finite number of cosets of U_E. $N_{F/E}^{-1}(U_E) = U_F$ which is compact (closed in a complete DVR). Thus the norm map is proper. A proper map is closed and the preimage of 0 is compact. So we have satisfied condition 1.
To find E_p in condition 2, we need multiple cases. If $p \neq char(K)$ then we simply take E_p to be the field obtained by adjoining the pth roots of unity to K. One can verify this indeed satisfies the condition of φ_p. If $p = char(K)$ then φ_p is the Frobenius automorphism on K. One can verify that $E_p = K$ works in this case. For the final condition, we take the compact subgroup needed to be U_E. Subgroups of finite index in E^* which contain U_E are preimages of the $n\mathbb{Z}$

under the valuation map. Chapter 13, section 4, proposition 13 in Serre shows that these are indeed norm groups.
We get the cleaner statement:

Theorem 19. Let K be a field complete under a discrete valuation with finite residue field. Then every closed subgroup of finite index in K^* is a norm group.

Corollary 11. The intersection of the norm groups is 1.

Corollary 12. Let $\mathfrak{I}_K$ is the inertia group of the maximal abelian extension K^{ab}/K then the reciprocity map $\theta : U_K \to \mathfrak{I}_K$ is an isomorphism of topological groups.

Corollary 13. $\mathrm{Gal}(K^{ab}/K)$ is an extension of $\hat{\mathbb{Z}}$ by U_K.

These can be found in chapter 11 of [10].

3.5 Hilbert Symbols

We now show how local class field theory applies to Hilbert Symbols. In the case $n = 2$, $K = \mathbb{Q}_p$ we have a beautiful and useful definition of the Hilbert symbol:

$$(a, b)_p = 1 \text{ if } z^2 = ax^2 + by^2 \text{ has a non-trivial solution over } \mathbb{Q}_p$$
$$(a, b)_p = -1 \text{ otherwise}$$

Using this and Hasse-Minkowski theorem will show when there are solutions to quadratics over $\mathbb{Q}$. We define the general case below:

Let K be a local field that contains a primitive n^{th} root of unity. Let μ_n be the group of nth roots of unity in K. Let K_s be the separable closure and G be the Galois group of K_s/K. There is a cup product pairing:

$$\mathrm{H}^1(G, \mu_n) \times \mathrm{H}^1(G, \mu_n) \to \mathrm{H}^2(G, \mu_n \otimes \mu_n)$$

However, we have some isomorphisms (Seen in Milne [13]) that make this look more familiar. $\mathrm{H}^1(G, K_s^*) \cong K^*/(K^*)^n$ and $\mathrm{H}^2(G, \mu_n) \cong \mu_n$. Therefore the cup product defines a pairing $K^*/(K^*)^n \times K^*/(K^*)^n \to \mu_n$. For $a, b \in K^*$, the image is denoted (a, b) and we call this the Hilbert Symbol. We state some results without proof to illustrate the importance of these symbols.

Theorem 20. The Hilbert Symbol satisfies:

1. $(aa', b) = (a, b)(a'b)$
2. $(a, bb') = (a, b)(a, b')$
3. $(b, a) = (a, b)^{-1}$
4. $(a, b) = 1$ for all $b \in K^*/(K^*)^n \Rightarrow a \in (K^*)^n$
5. $(a, b) = 1$ for all $a \in K^*/(K^*)^n \Rightarrow b \in (K^*)^n$
6. $(a, b) = 1$ if and only if b is a norm element from $K[a^{1/n}]$

The proof is found in Milne [13]. Hilbert symbols can also be used to prove quadratic reciprocity. Let $(a, b)_p$ denote the Hilbert symbol in the case $K = \mathbb{Q}_p$ and $(a, b)_\infty$ be the case $K = \mathbb{R}$.

Theorem 21 (Hilbert Product Formula [8]). $\prod (a,b)_p = 1$ as p runs over all primes and ∞.

We also have (seen in Cassels [14]):

$$(p,q)_r = \begin{cases} 1, & \text{if } r \text{ is positive odd and } r \neq p,q, \\ 1, & \text{if } r = \infty, \\ \left(\frac{p}{q}\right), & \text{if } r = q, \\ \left(\frac{q}{p}\right), & \text{if } r = p, \\ (-1)^{\frac{(p-1)(q-1)}{4}}, & \text{if } r = 2. \end{cases}$$

where $\left(\frac{p}{q}\right)$ is the Legendre symbol. Taking the product of these and applying the Hilbert product formula we get:

Theorem 22 (Quadratic Reciprocity). $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$

Proof. Hilbert product formula 21 and the above calculations give:

$$\begin{aligned} 1 &= \prod (a,b)_p \\ &= \left(\frac{p}{q}\right)\left(\frac{q}{p}\right)(-1)^{\frac{(p-1)(q-1)}{4}} \end{aligned}$$

Since the Legendre symbol only takes values $-1, 1$, we see that this implies that $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right)$ and $(-1)^{\frac{(p-1)(q-1)}{4}}$ always have the same sign and hence are always equal. □

These results are all very beautiful, but how does this relate to class field theory? The Hilbert symbol can be related to the Artin map via the following identity [13]:

$$\theta_{L/K}(b) = (a,b)\, a^{\frac{1}{n}}.$$

Where K again is a field containing the nth roots of unity and $L = K[a^{\frac{1}{n}}]$. We have proven that this map is an isomorphism between K^*/NL^* and $\mathrm{Gal}(L/K)$. This Galois group is cyclic of order n and isomorphic to μ_n and $NL^* = (K^*)^n$. Hence the Hilbert symbol, being a root of unity itself, acts on the Galois group μ_n in the natural way. Therefore, if we can compute the image of b under the reciprocity isomorphism, we can deduce the Hilbert symbol and vice versa. The Hilbert symbol becomes an explicit way to realize how elements of K^* act on abelian extensions, giving one a powerful computational tool in local class field theory. We work through an example to illustrate:

Example 8. We take $n = 2$. Let p be an odd prime. Let $a = u \cdot p^\alpha$, $b = v \cdot p^\beta$ for u, v units and $\alpha, \beta \in \mathbb{Z}$. Cassels proves the explicit formula $(a,b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{u}{p}\right)^\beta \left(\frac{v}{p}\right)^\alpha$ in this case. If we take $p = 5$ and compute $(5,2)$ we get -1, meaning that 2 is not a norm element in the extension $\mathbb{Q}_5[\sqrt{5}]/\mathbb{Q}_5$. Its coset in $(\mathbb{Q}_5^*)/N(\mathbb{Q}_5[\sqrt{5}]^*)$ must map to a non-trivial element in the Galois group of that extension.

3.6 Examples

We now apply the theorems to see how they work.

Example 9. Let L be an unramified abelian extension of $\mathbb{Q}_p$ of degree n. Let $x \in L^*$. Then $x = p^m \cdot u$ for some unit $u \in U_L$ and $m \in \mathbb{Z}$ (since $L/\mathbb{Q}_p$ is unramified). Then the image of x under the norm map is $p^{nm} \cdot u'$ for some unit in U_K (as we have seen the image of U_L under the norm is U_K). Then we have $N_{L/\mathbb{Q}_p}(L^*) = \langle p^n \rangle \cdot U_K$. We see:

$$\begin{aligned}\mathbb{Q}_p^*/N_{L/\mathbb{Q}_p}(L^*) =& (\langle p \rangle \cdot U_{\mathbb{Q}_p})/(\langle p^n \rangle \cdot U_{\mathbb{Q}_p}) \\ =& \mathbb{Z}/n\mathbb{Z}\end{aligned}$$

By the reciprocity isomorphism, $\mathrm{Gal}(L/\mathbb{Q}_p) = \mathbb{Z}/n\mathbb{Z}$. In addition, by the norm group computation we see that $\langle p^n \rangle \cdot U_{\mathbb{Q}_p}$ is the only subgroup of $\mathbb{Q}_p^* = \langle p \rangle \cdot U_{\mathbb{Q}_p}$ of order n which is also a norm group of an unramified extension. We see that there is a unique unramified extension of $\mathbb{Q}_p$ with degree n.

Example 10. Let $K = \mathbb{Q}_p$ and L be the field obtained by adjoining ζ, a $(p^n)^{th}$ root of unity. Then the Galois group is $(\mathbb{Z}/p^n\mathbb{Z})^*$ since any automorphism sends ζ to ζ^a for some a such that $(a, p) = 1$. Then this is the totally ramified extension of degree $\varphi(p^n) = p^{n-1}(p-1)$.

Example 11. This example is due to Serre [1]. Let $K = \mathbb{Q}_p$ and K^{ab} be the maximal abelian extension of K. Then K^{ab} is the extension E obtained by adjoining all roots of unity to K. Indeed, E is the disjoint composite extension of the maximal unramified extension K_{nr} and the maximal totally ramified extension K_{tr}. It is abelian, so it is a subextension of K^{ab}. From the corollary above we deduce the isomorphism:

$$\hat{\mathbb{Z}} \cdot U_K \longrightarrow \mathrm{Gal}(K^{ab}/K)$$

We have also seen that $\mathrm{Gal}(K_{nr}/K) = \hat{\mathbb{Z}}$ so we conclude $\mathrm{Gal}(K^{ab}/K_{nr}) = U_K$. Now $\mathrm{Gal}(E/K_{nr}) = \mathrm{Gal}(K_{tr}/K)$ which is given by the limit $\varprojlim \mathrm{Gal}(K(\zeta_{p^n})/K) = \varprojlim (\mathbb{Z}/p^n\mathbb{Z})^*$ (by the example above). One can see this is $\mathbb{Z}_p^*$ or U_K. Hence we have a surjective homomorphism $\mathrm{Gal}(K^{ab}/K_{nr}) \to \mathrm{Gal}(E/K_{nr})$ which is from U_K to itself. However, U_K is a finitely generated $\hat{\mathbb{Z}}$-module and any surjective endomorphism of such a module is bijective [6].

Corollary 14 (Local Kronecker-Weber)**.** Any abelian extension of $\mathbb{Q}_p$ is contained in some cyclotomic extension.

Proof. This follows since, in the notation of the example above, any finite subextension of K^{ab} is contained in some finite subextension of E. This in turn is contained in some appropriate finite cyclotomic extension. □

Example 12. Let's find the degree 5 abelian extensions of $\mathbb{Q}_p$, $p \neq 5$. We write $\mathbb{Q}_p^* = p^{\mathbb{Z}} \cdot \mu_{p-1} \cdot U_{\mathbb{Q}_p}^{(1)}$. We now look for order 5 subgroups of $\mathbb{Q}_p^*/(\mathbb{Q}_p^*)^5$. 5 is a unit in $\mathbb{Z}_p$ so $(U_{\mathbb{Q}_p}^{(1)})^5 = U_{\mathbb{Q}_p}^{(1)}$ and this contributes nothing to the quotient. $p^{\mathbb{Z}}/p^{5\mathbb{Z}}$ contributes $\mathbb{Z}/5\mathbb{Z}$. If $5|(p-1)$ then $(\mu_{p-1})^5 \neq \mu_{p-1}$ and is an index 5 subgroup of μ_{p-1}. Then $\mu_{p-1}/(\mu_{p-1})^5$ contributes a $\mathbb{Z}/5\mathbb{Z}$. Otherwise, $(\mu_{p-1})^5 = \mu_{p-1}$ and this contributes nothing.

Hence if $5 \nmid (p-1)$ then $\mathbb{Q}_p^*/(\mathbb{Q}_p^*)^5 = \mathbb{Z}/5\mathbb{Z}$ and we can have only one extension of degree 5 (the unique unramified extension by the example above).

If $5|(p-1)$ then we have $\mathbb{Q}_p^*/(\mathbb{Q}_p^*)^5 = (\mathbb{Z}/5\mathbb{Z})^2$. The number of subgroups of this of order 5 is $24/4 = 6$ (number of elements of order 5 divided by number of non-identity elements in

each subgroup. Hence in this case we have 6 abelian extensions of $\mathbb{Q}_p$. One is unramified and the other 5 are totally tamely ramified, hence equal to $\mathbb{Q}_p[\alpha]$ for α some root of an Eisenstein polynomial by proposition 3.

Indeed we have the more general result:

Example 13. Let $p \neq 2$. There are $p+1$ degree-p extensions of $\mathbb{Q}_p$. Via analogous calculations to the above example, we have that $\mathbb{Q}_p^*/(\mathbb{Q}_p^*)^p \cong (\mathbb{Z}/p\mathbb{Z})^2$. Then the number of subgroups of order p in here are $\frac{p^2-1}{p-1} = p+1$.

Example 14. There are 7 degree-2 extensions of $\mathbb{Q}_2$. Cassels shows $(\mathbb{Q}_2^*) \cong \mathbb{Z} \cdot \mathbb{Z}_2 \cdot \mathbb{Z}/2\mathbb{Z}$ [15]. Then we see the quotient $(\mathbb{Q}_2^*)/(\mathbb{Q}_2^*)^2 \cong (\mathbb{Z}/2\mathbb{Z})^3$. There are $2^3 - 1$ order 2 elements here, each generating an order 2 subgroup. This gives us the 7 extensions we want.

With that example and the one from the introduction we have found the quadratic extensions of any $\mathbb{Q}_p$.

Chapter 4

Conclusion

Class field theory as a whole was a major mathematical achievement of the 20th century. The cohomological approach seen here was developed after the theory was originally discovered. It provides a clean avenue to the main theorems and, in the opinion of the author, is beautiful and satisfying to work through.

The natural extension of this theory is to look towards global class field theory. Indeed, one can use a local to global approach and continue with the cohomological methods to do so. Understanding field extensions of number fields is of great importance in number theory and the applications are endless. Future students may find this report to be a jumping off point to explore the ideas and proofs in global class field theory.

The work shown here has gone through great detail to understand the proofs of local Artin reciprocity and the existence theorem essentially in their complete form. Indeed with these theorems we were able to establish a 1-to-1 correspondence between abelian extension of a local field K and closed subgroups of K^* with finite index. In addition, we built an understanding of Galois cohomology which has many uses in arithmetic geometry and number theory.

In conclusion, I thank the reader for their attention. I hope the proofs and exposition were as rewarding to read as they were to write.

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